

The Joy of Pseudo-Boolean Proof Logging

Jakob Nordström

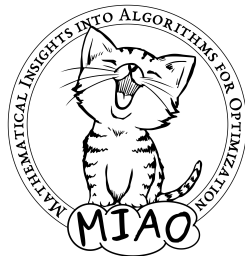
University of Copenhagen and Lund University

ACP Summer School 2026

CP: Explanation, Verification, and Applications

Changchun, China

August 24–28, 2026



The Joy of Pseudo-Boolean Proof Logging

Or: An Introduction to VERIPB

Jakob Nordström

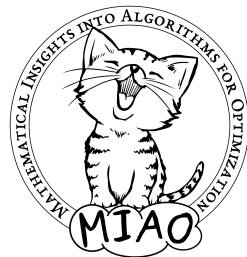
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Brief Presentation

Jakob Nordström

University of Copenhagen
and Lund University

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www.jakobnordstrom.se

Research interests:

- Computational complexity theory
- Applied combinatorial optimization
- Certifying algorithms / proof logging



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Research interests:

- *Proving exponential lower bounds for NP-hard problems*
- *Designing algorithms to solve NP-hard problems in (ideally) linear time*
- *Generating machine-verifiable proofs that such algorithms reason correctly*



Tutorial Based on Research Contributions By...

■ Berhan Oumer Adame

■ Markus Anders

■ Jeremias Berg

■ Bart Bogaerts

■ Benjamin Bogø

■ Wolf De Wulf

■ Emir Demirović

■ Simon Dold

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Combinatorial Solving and Optimisation

- Revolution last couple of decades in **combinatorial solvers** for
 - Boolean satisfiability (SAT) solving [BHvMW21]¹
 - Constraint programming (CP) [RvBW06]
 - Mixed integer linear programming (MIP) [AW13, BR07]
- Solve NP-complete problems (or worse) very successfully in practice!
- **Except solvers are sometimes wrong...** (Even best commercial ones) [BLB10, CKSW13, AGJ⁺18, GSD19, BMN22, GCS23]
- Solvers can propose infeasible “solutions” (but such errors can be detected)
- More challenging: How to achieve reliable claims of infeasibility?
- Or that a solution is optimal? (Even off-by-one mistakes can snowball into large errors if solver used as subroutine)

¹See end of slides for all references with bibliographic details

What Can Be Done About Solver Bugs?

■ Software testing

Very useful, but bugs slip through even with careful domain-specific testing

Progress using [fuzzing](#) and [delta debugging](#) [BB09, BLB10, KB22, NPB22, PB23]

But testing inherently can only detect presence of bugs, not absence

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■ Formal methods

Prove that solver implementation adheres to formal specification

Current [formal verification](#) techniques cannot scale to level of complexity in modern solvers

(Despite valiant efforts in, e.g., [Fle20])

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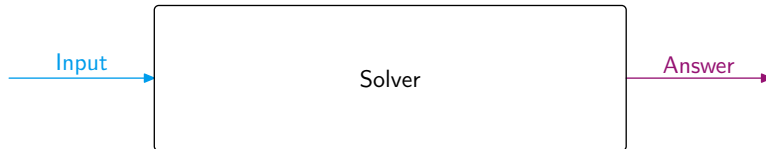
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■ Proof logging

Make solver [certifying](#) [ABM⁺11, MMNS11] by adding code so that it outputs

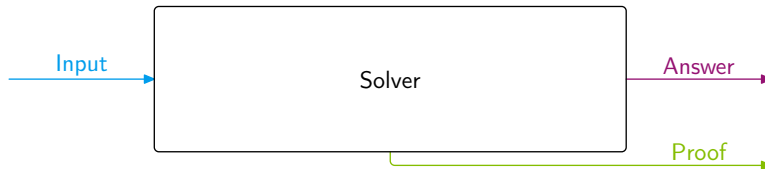
- 1 not only [answer](#) but also
- 2 simple, machine-verifiable [proof](#) that answer is correct

Proof Logging with Certifying Solvers: Workflow



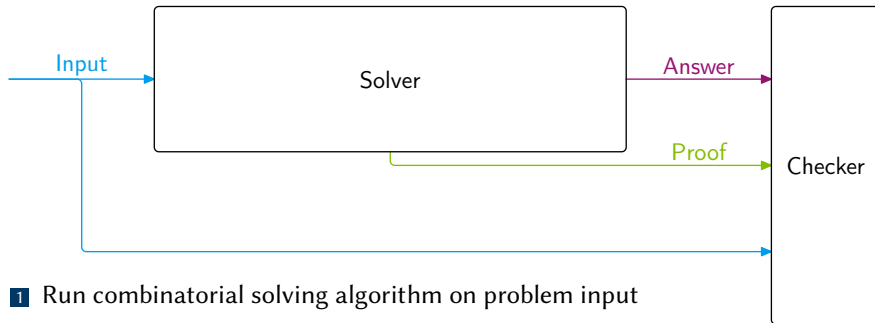
- 1 Run combinatorial solving algorithm on problem input

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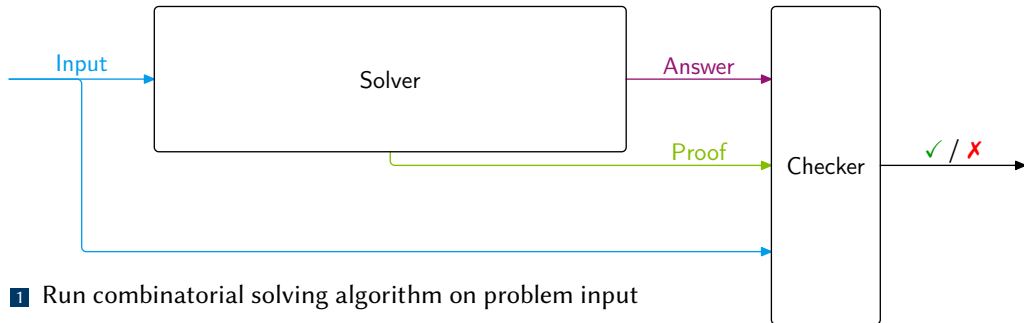
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Proof Logging with Certifying Solvers: Workflow



- 1 Run combinatorial solving algorithm on problem input
- 2 Get as output not only answer but also proof
- 3 Feed input + answer + proof to proof checker

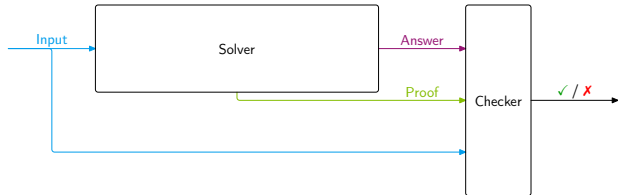
Proof Logging with Certifying Solvers: Workflow



- 1 Run combinatorial solving algorithm on problem input
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- 4 Verify that proof checker says answer is correct

Proof Logging Desiderata

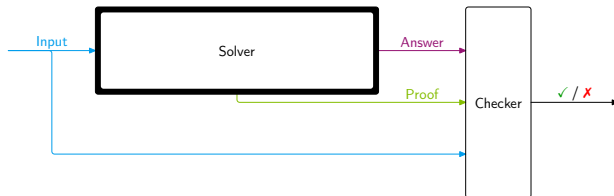
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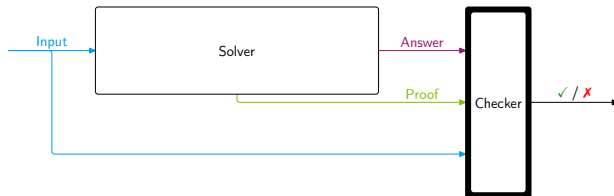
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Proof Logging Desiderata

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- **very powerful**: minimal overhead for sophisticated reasoning
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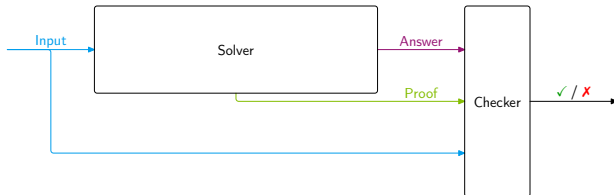


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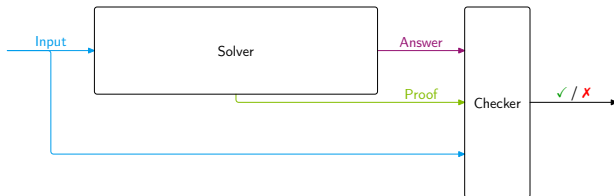
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Clear conflict expressivity vs. simplicity!



Proof Logging Desiderata



Proof format for certifying solver should be

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Clear conflict expressivity vs. simplicity!

Asking for both perhaps a little bit too good to be true?

Some Previous Proof Logging Work

Boolean satisfiability (SAT) solving

- Well established since over decade with several proof formats such as [DRAT](#) [HHW13a, HHW13b, WHH14], [GRIT](#) [CMS17], [LRAT](#) [CHH⁺17], ...
- But no efficient support for most advanced techniques such as
 - Gaussian elimination
 - symmetry breaking
- And only for decision problems, not optimization (MaxSAT)

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Constraint programming

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Mixed integer linear programming

- Work on proof format [VIPR](#) [CGS17, EG23]
- But only for exact solving and without support for advanced techniques

Take-Away Message from This Tutorial

Proof logging for sophisticated combinatorial solvers is possible!



Take-Away Message from This Tutorial

Proof logging for sophisticated combinatorial solvers is possible!

- With **single, unified method!**
- Producing proofs in simple format calculating with **0–1 integer linear inequalities** (a.k.a. **pseudo-Boolean constraints**)
- Implemented in **VERIPB** (<https://VeriPB.org>)



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- 1 Marketing pitch 😊



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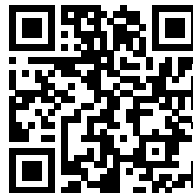
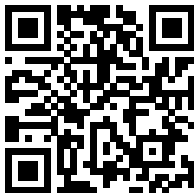
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These slides with references are online at <https://jakobnordstrom.se/presentations/>

Software for Exercises Tomorrow Thursday



Kindling Solver: See [README.md](#) and files in [slides-proofs/](#)

<https://github.com/ciaranm/kindling>

VeriPB (official version): See [README.md](#) and try to run `veripb justified.opb justified.pb`

<https://gitlab.com/MIA0research/software/VeriPB>

VeriPB (experimental): Interactive mode (Linux and Mac only?); see [README.md](#) and then use `cargo run -p veripb-repl -m path-to-veripb-repl/Cargo.toml`

<https://github.com/ciaranm/veripb-repl>

The Sales Pitch For Proof Logging

- 1 Certifies correctness of computed results

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- 2 **Detects errors** even if due to compiler bugs, hardware failures, or cosmic rays

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- 6 Enables auditability
- 7 Serves as stepping stone towards explainability

Design Principles for Proof Logging

Proof logging implementation

- Don't change solver
- Just add proof logging print statements (plus some book-keeping) to solver code

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- Proof logging overhead small constant fraction of running time ($\lesssim 10\%$)
- Proof checking time within constant factor of solving time (current aim $\lesssim \times 10$)

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Proof system

- Keep language simple — no XOR constraints, CP propagators, symmetry groups, ...
- But reason efficiently about such notions using power of proof system
- Combine proof logging with formally verified proof checker

But Let Us Start from the Beginning...

Review of some basic concepts:

- SATISFIABILITY (SAT) problem
- Unit propagation
- DPLL and CDCL algorithms
- Resolution proof system

The SATISFIABILITY (SAT) Problem

- **Variable** x : takes value **true** (=1) or **false** (=0)
- **Literal** ℓ : variable x or its negation $\bar{x}$
- **Clause** $C = \ell_1 \vee \dots \vee \ell_k$: disjunction of literals
(Consider as sets, so no repetitions and order irrelevant)
- **Conjunctive normal form (CNF) formula** $F = C_1 \wedge \dots \wedge C_m$: conjunction of clauses

The SAT Problem

Given a CNF formula F , is it satisfiable?

For instance, what about:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge \\ (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

Proofs for SAT

For satisfiable instances: just specify satisfying assignment

For unsatisfiability: a sequence of clauses (CNF constraints)

- Each clause follows “obviously” from everything we know so far
- Final clause is empty, meaning contradiction (written $\perp$)
- Means original formula must be inconsistent

What Is Obvious? Unit Propagation

Unit Propagation

Clause C **unit propagates** ℓ under partial assignment ρ if ρ falsifies all literals in C except ℓ

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- $p \vee \bar{u}$ propagates $u \mapsto 0$

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- No further unit propagations

Proof checker should know how to unit propagate until saturation

Davis-Putman-Logemann-Loveland (DPLL)

DPLL [DP60, DLL62]: Assign variables and propagate; backtrack when clause violated

“Proof trace”: when backtracking, write negation of guesses made

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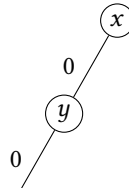


Davis-Putman-Logemann-Loveland (DPLL)

DPLL [DP60, DLL62]: Assign variables and propagate; backtrack when clause violated

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$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee \textcolor{violet}{x} \vee \textcolor{violet}{y}) \wedge (\textcolor{violet}{x} \vee \textcolor{teal}{y} \vee z) \wedge (\bar{\textcolor{teal}{x}} \vee z) \wedge (\textcolor{teal}{y} \vee \bar{z}) \wedge (\bar{\textcolor{teal}{x}} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

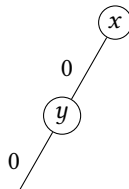


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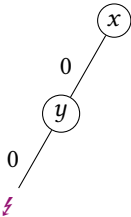
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1 $x \vee y$



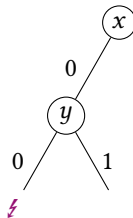
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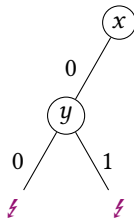
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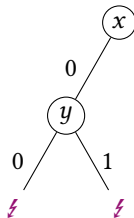
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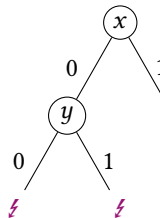
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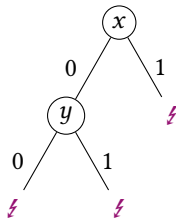
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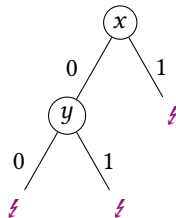
1 $x \vee y$

2 $x \vee \bar{y}$

3 x

4 $\bar{x}$

5 $\perp$



Reverse Unit Propagation (RUP)

To make this a proof, need backtrack clauses to be easily verifiable

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- assigning C to false
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Fact

Backtrack clauses from DPLL solver generate a RUP proof

What About Conflict-Driven Clause Learning (CDCL)?

Run CDCL [BS97, MS99, MMZ⁺01] on our favourite CNF formula:

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Free choice to assign value to variable

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Unit propagation

Forced choice to avoid falsifying clause

Given $p = 0$, clause $p \vee \bar{u}$ forces $u = 0$

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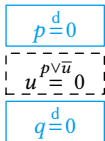
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Continue until satisfying assignment or **conflict**

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$$q \stackrel{d}{=} 0$$

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$$x \stackrel{d}{=} 0$$

$$y \stackrel{u \vee x \vee y}{=} 1$$

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$$z \stackrel{x \vee \bar{y} \vee z}{=} 1$$

Decision

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Unit propagation

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$$x \stackrel{d}{=} 0$$

$$y \stackrel{u \vee x \vee y}{=} 1$$

$$z \stackrel{x \vee \bar{y} \vee z}{=} 1$$

$$\bar{y} \vee \bar{z} \stackrel{\perp}{=}$$

Decision

Free choice to assign value to variable

Notation $p \stackrel{d}{=} 0$

Unit propagation

Forced choice to avoid falsifying clause

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$$p \stackrel{d}{=} 0$$

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$$y \stackrel{u \vee x \vee y}{=} 1$$

$$z \stackrel{x \vee \bar{y} \vee z}{=} 1$$

$$\bar{y} \vee \bar{z}$$

decision
level 1

Decision

Free choice to assign value to variable

Notation $p \stackrel{d}{=} 0$

decision
level 2

Unit propagation

Forced choice to avoid falsifying clause

Given $p = 0$, clause $p \vee \bar{u}$ forces $u = 0$

Notation $u \stackrel{p \vee \bar{u}}{=} 0$ ($p \vee \bar{u}$ is **reason clause**)

decision
level 3

Always propagate if possible, otherwise decide

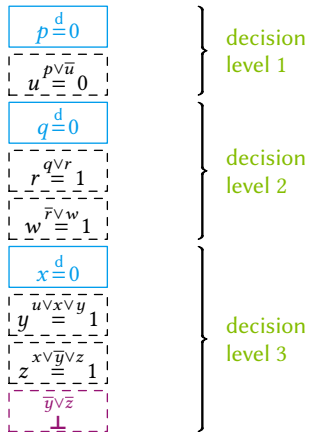
Add to assignment **trail**

Continue until satisfying assignment or **conflict**

Conflict Analysis

Time to analyse this conflict and learn from it!

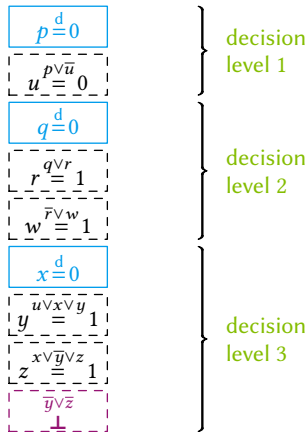
$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



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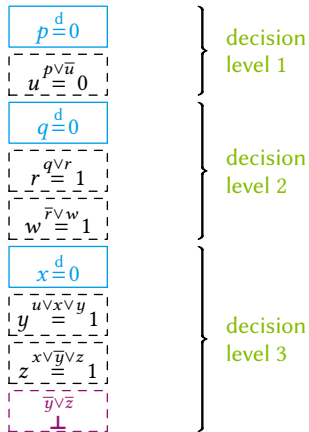


Could backtrack by erasing **conflict level** & flipping last decision

Conflict Analysis

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$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



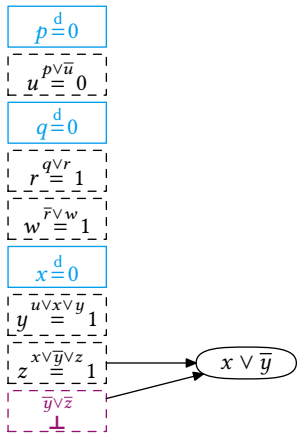
Could backtrack by erasing **conflict level** & flipping last decision

But want to **learn** from conflict and cut away as much of search space as possible

Conflict Analysis

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$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$



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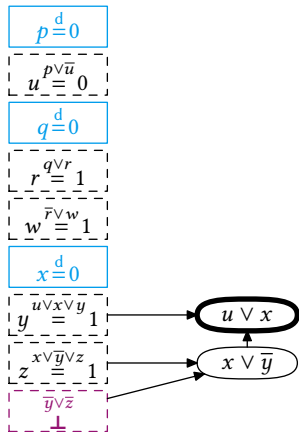
Case analysis over z for last two clauses:

- $x \vee \bar{y} \vee z$ wants $z = 1$
- $\bar{y} \vee \bar{z}$ wants $z = 0$
- **Resolve** clauses by merging them & removing z — must satisfy $x \vee \bar{y}$

Conflict Analysis

Time to analyse this conflict and learn from it!

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



Could backtrack by erasing **conflict level** & flipping last decision

But want to **learn** from conflict and cut away as much of search space as possible

Case analysis over z for last two clauses:

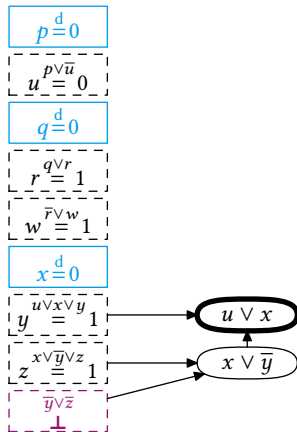
- $x \vee \bar{y} \vee z$ wants $z = 1$
- $\bar{y} \vee \bar{z}$ wants $z = 0$
- **Resolve** clauses by merging them & removing z — must satisfy $x \vee \bar{y}$

Repeat until **UIP clause** with only 1 variable at conflict level after last decision — **learn** and **backjump**

Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



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$$p \stackrel{d}{=} 0$$

$$u \stackrel{p \vee \bar{u}}{=} 0$$

$$q \stackrel{d}{=} 0$$

$$r \stackrel{q \vee r}{=} 1$$

$$w \stackrel{\bar{r} \vee w}{=} 1$$

$$x \stackrel{d}{=} 0$$

$$y \stackrel{u \vee x \vee y}{=} 1$$

$$z \stackrel{x \vee \bar{y} \vee z}{=} 1$$

$$\bar{y} \vee \bar{z}$$

$$\perp$$

$$u \vee x$$

$$x \vee \bar{y}$$

$$p \stackrel{d}{=} 0$$

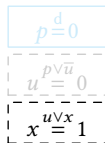
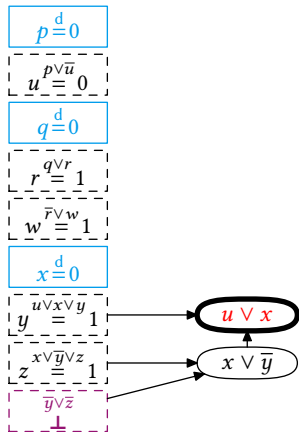
$$u \stackrel{p \vee \bar{u}}{=} 0$$

Assertion level 1 (2nd largest level in learned clause) — trim trail to that level

Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

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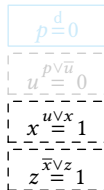
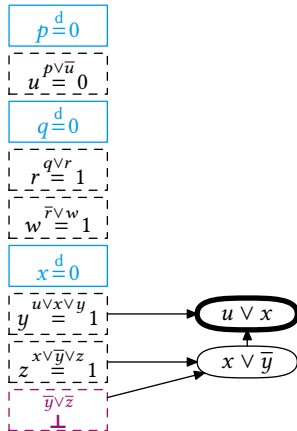
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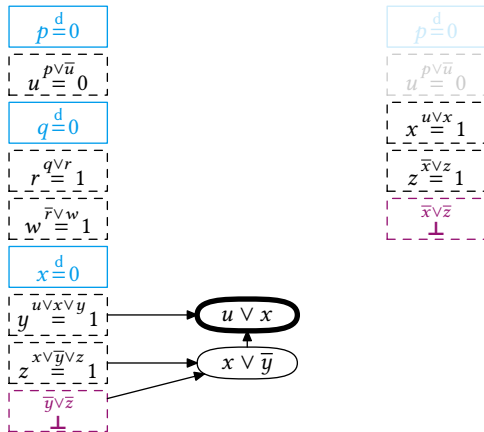
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Then continue as before...

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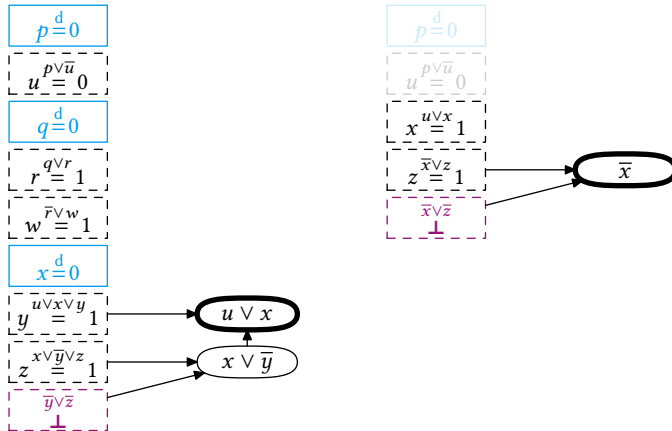
$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$



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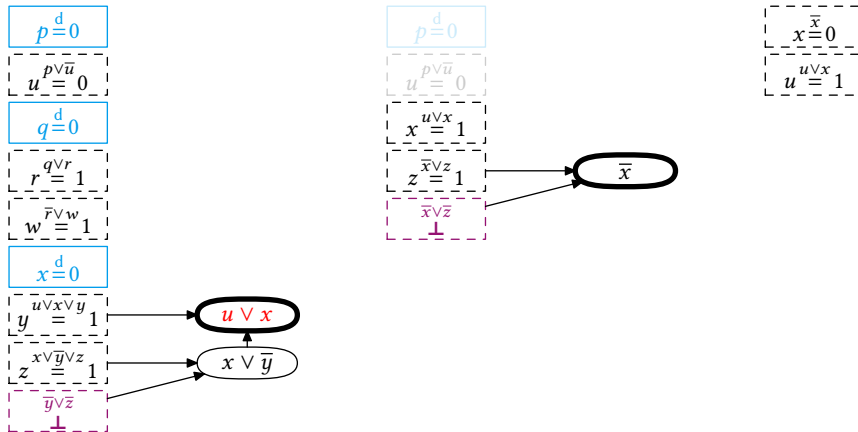
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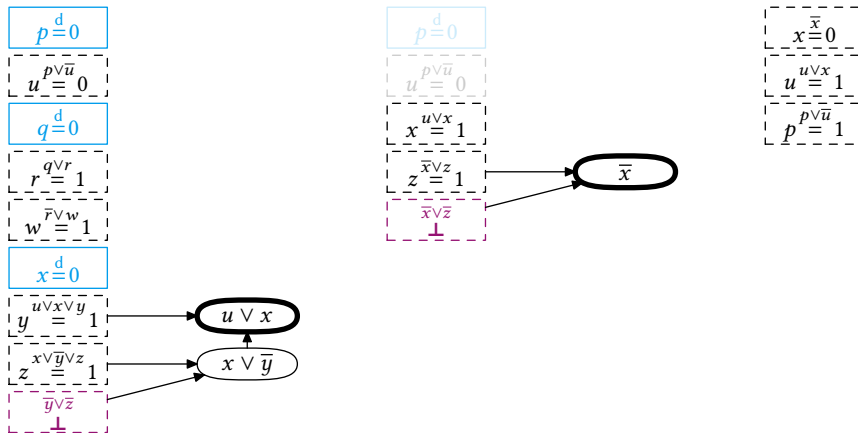
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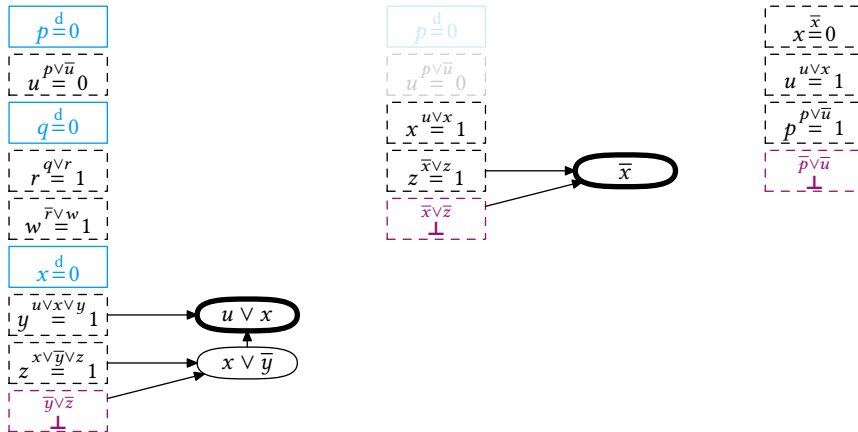
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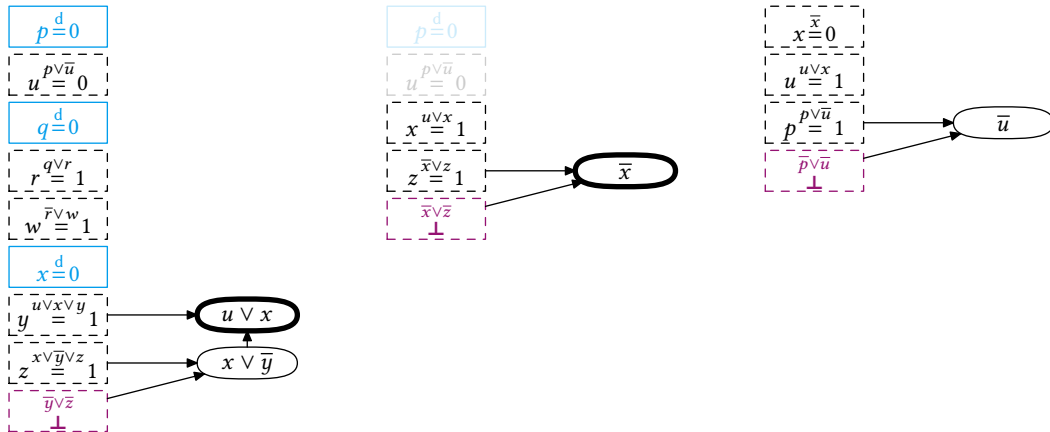
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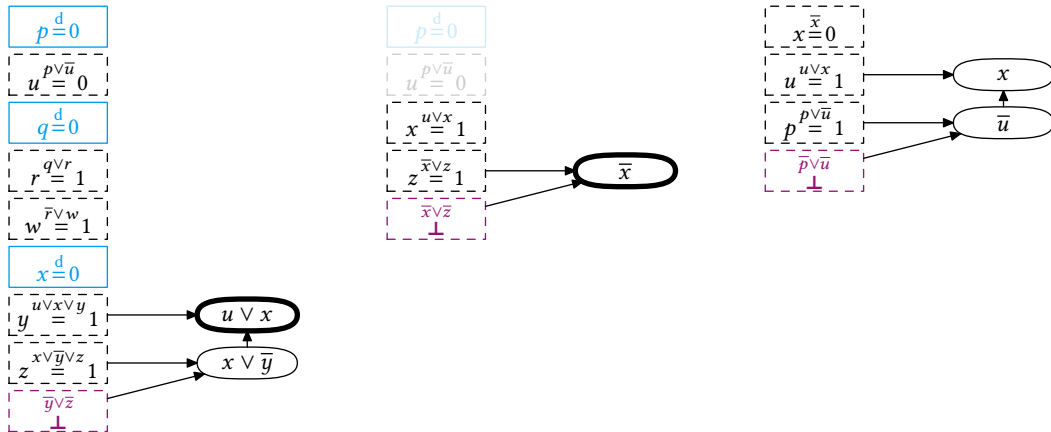
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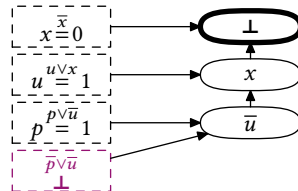
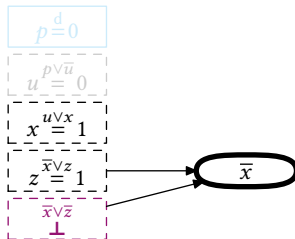
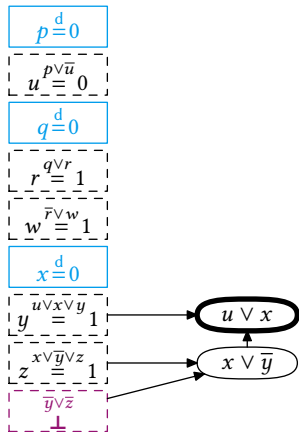
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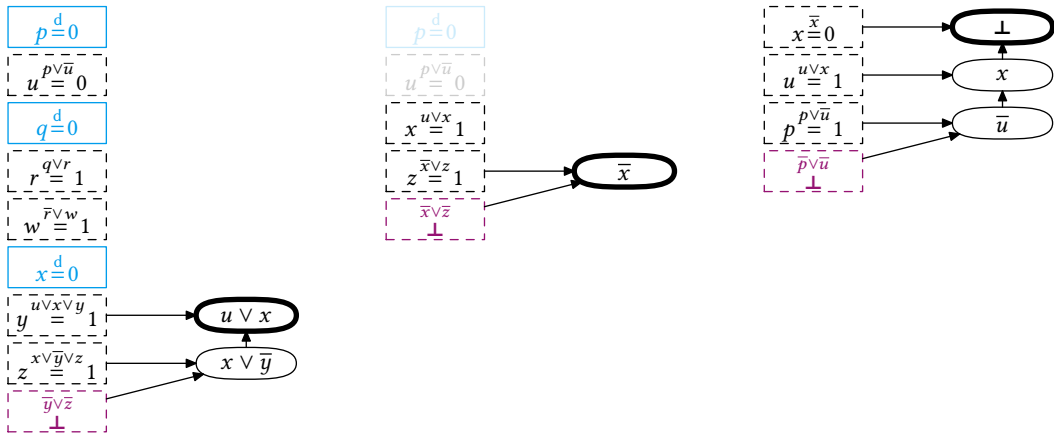
(*) Ignores pre- and inprocessing, but we will get there...

Resolution Proofs from CDCL Executions

Obtain resolution proof...

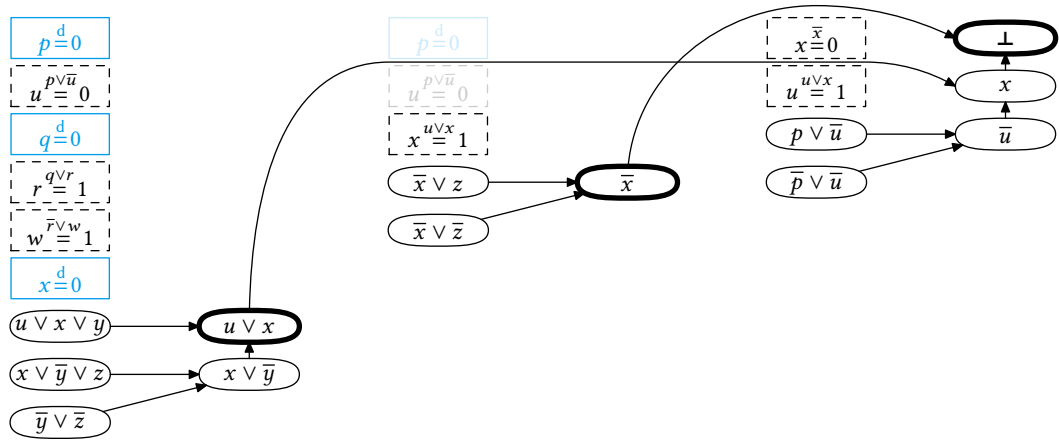
Resolution Proofs from CDCL Executions

Obtain resolution proof from our example CDCL execution...



Resolution Proofs from CDCL Executions

Obtain resolution proof from our example CDCL execution by stringing together conflict analyses:



RUP Proofs and CDCL

But it turns out we can be lazier...

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More Ingredients in Proof Logging for SAT

Fact

RUP proofs can be viewed as shorthand for resolution proofs

See proof complexity and SAT solving survey [BN21] for more on this

But RUP and resolution are not enough for preprocessing, inprocessing, and some other kinds of reasoning

Extension Variables, Part 1

Suppose we want a **reified variable** r encoding

$$r \Leftrightarrow (x \wedge y)$$

Extended resolution [Tse68]

Resolution rule plus **extension rule** introducing clauses

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Fact

Extended resolution (RUP + definition of new variables) is essentially equivalent to the DRAT proof logging system most commonly used for SAT solving [KRH18]

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Practical limitations of current SAT proof logging technology:

- Difficulties dealing with stronger reasoning efficiently (even for SAT solving)
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- And more...

Pseudo-Boolean Constraints

0–1 integer linear inequalities or (linear) pseudo-Boolean constraints:

$$\sum_i a_i \ell_i \geq A$$

- $a_i, A \in \mathbb{Z}$
- **literals** ℓ_i : x_i or $\bar{x}_i$ (where $x_i + \bar{x}_i = 1$)

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Sometimes convenient to use **normalised form** [Bar95] with **all a_i, A positive** (without loss of generality)

Some Types of Pseudo-Boolean Constraints

1 Clauses

$$x_1 \vee \bar{x}_2 \vee x_3 \quad \Leftrightarrow \quad x_1 + \bar{x}_2 + x_3 \geq 1$$

2 Cardinality constraints

$$x_1 + x_2 + x_3 + x_4 \geq 2$$

3 General pseudo-Boolean constraints

$$x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$$

Pseudo-Boolean Reasoning: Cutting Planes [CCT87]

Input/model axioms

From the input

Pseudo-Boolean Reasoning: Cutting Planes [CCT87]

Input/model axioms

Literal axioms

From the input

$$\overline{\ell_i \geq 0}$$

Pseudo-Boolean Reasoning: Cutting Planes [CCT87]

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Addition

From the input

$$\overline{\ell_i \geq 0}$$

$$\frac{\sum_i a_i \ell_i \geq A \quad \sum_i b_i \ell_i \geq B}{\sum_i (a_i + b_i) \ell_i \geq A + B}$$

Pseudo-Boolean Reasoning: Cutting Planes [CCT87]

Input/model axioms

From the input

Literal axioms

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Addition

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Multiplication for any $c \in \mathbb{N}^+$

$$\frac{\sum_i a_i \ell_i \geq A}{\sum_i c a_i \ell_i \geq cA}$$

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$$\frac{\sum_i a_i \ell_i \geq A}{\sum_i \lceil \frac{a_i}{c} \rceil \ell_i \geq \lceil \frac{A}{c} \rceil}$$

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$$\overline{\ell_i \geq 0}$$

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Saturation for any $c \in \mathbb{N}^+$
(assumes normalised form)

$$\frac{\sum_i a_i \ell_i \geq A}{\sum_i \min(a_i, A) \cdot \ell_i \geq A}$$

Cutting Planes Toy Example

$$w + 2x + y \geq 2$$

Cutting Planes Toy Example

$$\text{Multiply by 2} \quad \frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4}$$

Cutting Planes Toy Example

$$\text{Multiply by 2} \quad \frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4} \quad w + 2x + 4y + 2z \geq 5$$

Cutting Planes Toy Example

$$\begin{array}{rcl}
 & w + 2x + y \geq 2 & \\
 \text{Multiply by 2} & \frac{2w + 4x + 2y \geq 4}{\text{Add}} & w + 2x + 4y + 2z \geq 5 \\
 & \hline
 & 3w + 6x + 6y + 2z \geq 9 &
 \end{array}$$

Cutting Planes Toy Example

$$\begin{array}{rcl}
 \text{Multiply by 2} & \frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4} & \\
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 \end{array}$$

Cutting Planes Toy Example

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 & & \frac{\bar{z} \geq 0}{2\bar{z} \geq 0} \text{ Multiply by 2}
 \end{array}$$

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 \text{Add} & \frac{\bar{z} \geq 0}{2\bar{z} \geq 0} & \text{Multiply by 2} \\
 & \frac{3w + 6x + 6y + 2z \geq 9}{3w + 6x + 6y + 2z + 2\bar{z} \geq 9} &
 \end{array}$$

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$$\begin{array}{rcl} \text{Multiply by 2} & \frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4} & \\ \text{Add} & \frac{w + 2x + 4y + 2z \geq 5}{3w + 6x + 6y + 2z \geq 9} & \\ & \text{Add} & \\ & \frac{2\bar{z} \geq 0}{2\bar{z} \geq 0} & \text{Multiply by 2} \\ & \frac{3w + 6x + 6y + 2z \geq 9}{3w + 6x + 6y + 2} & \geq 9 \end{array}$$

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 \text{Add} & \frac{3w + 6x + 6y + 2z \geq 9}{3w + 6x + 6y} & \geq 7
 \end{array}$$

Cutting Planes Toy Example

$$\begin{array}{rcl}
 \text{Multiply by 2} & \frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4} & \\
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 \text{Add} & \frac{3w + 6x + 6y + 2z \geq 9}{3w + 6x + 6y} & \geq 7 \\
 \text{Divide by 3} & \frac{3w + 6x + 6y}{w + 2x + 2y} & \geq 2\frac{1}{3}
 \end{array}$$

Cutting Planes Toy Example

$$\begin{array}{rcl}
 & w + 2x + y \geq 2 & \\
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 & \frac{3w + 6x + 6y + 2z \geq 9}{\text{Divide by 3}} & \frac{2\bar{z} \geq 0}{\geq 7} \\
 & \frac{w + 2x + 2y \geq 3}{\geq 3} &
 \end{array}$$

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Naming constraints by integers and literal axioms by the literal involved (with $\sim$ for negation) as

$$\text{Constraint 1} \doteq 2x + y + w \geq 2$$

$$\text{Constraint 2} \doteq 2x + 4y + 2z + w \geq 5$$

$$\sim z \doteq \bar{z} \geq 0$$

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such a calculation is written in the proof log in reverse Polish notation as*

pol 1 2 * 2 + $\sim z$ 2 * + 3 d ;

Cutting Planes Toy Example

Multiply by 2

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Add

$$\frac{w + 2x + 4y + 2z \geq 5}{3w + 6x + 6y + 2z \geq 9}$$

Add

$$\frac{\bar{z} \geq 0}{2\bar{z} \geq 0}$$

Multiply by 2

$$\frac{2\bar{z} \geq 0}{2\bar{z} \geq 0}$$

$$\frac{3w + 6x + 6y + 2z \geq 9}{3w + 6x + 6y} \geq 7$$

Divide by 3

$$\frac{3w + 6x + 6y}{w + 2x + 2y} \geq 3$$

Naming constraints by integers and literal axioms by the literal involved (with ~ for negation) as

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Constraint 2

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~z

$\doteq \bar{z} \geq 0$

such a calculation is written in the proof log in reverse Polish notation as*

pol

1

2

*

2

+

~z

2

*

+

3

d

;

(*) Except VERIPB variables have to be at least two characters long, but we abuse syntax slightly here for simplicity

Resolution and Cutting Planes

To simulate resolution step such as

$$\frac{\bar{y} \vee \bar{z} \quad x \vee \bar{y} \vee z}{x \vee \bar{y}}$$

we can perform the cutting planes steps

$$\begin{array}{l} \text{Add} \quad \frac{\bar{y} + \bar{z} \geq 1 \quad x + \bar{y} + z \geq 1}{x + 2\bar{y} \geq 1} \\ \text{Divide by 2} \quad \frac{x + 2\bar{y} \geq 1}{x + \bar{y} \geq 1} \end{array}$$

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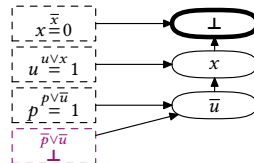
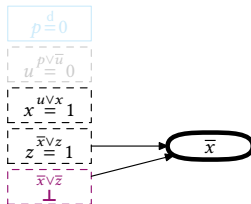
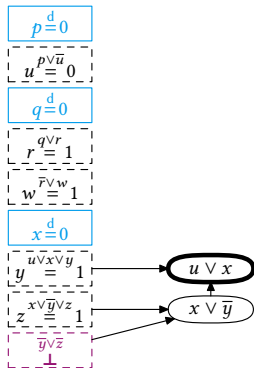
Given that the premises are clauses 7 and 5 in our example CNF formula, using references

$$\begin{array}{l} \text{Constraint 7} \doteq \overline{y} + \overline{z} \geq 1 \\ \text{Constraint 5} \doteq x + \overline{y} + z \geq 1 \end{array}$$

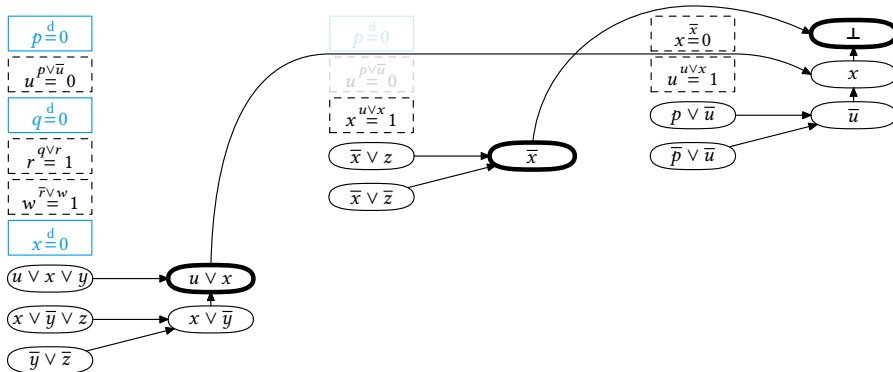
we can write this in the proof log as

$$\text{pol } 7 \ 5 \ + \ 2 \ d \ ;$$

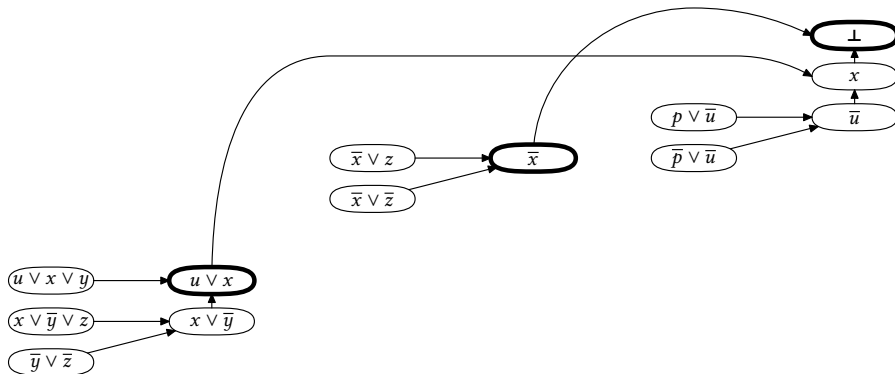
Pseudo-Boolean Proof Logging for Example CDCL Conflict Analyses



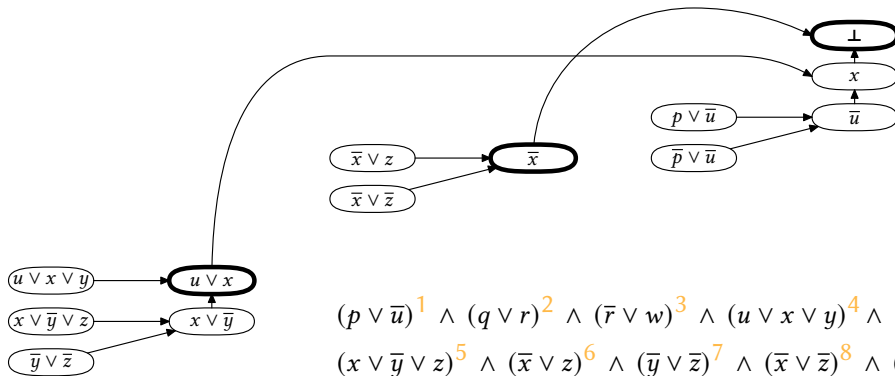
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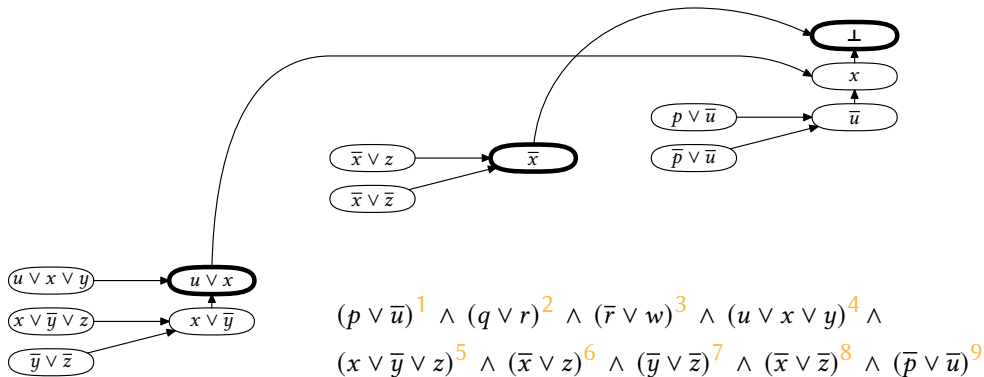
Pseudo-Boolean Proof Logging for Example CDCL Conflict Analyses



Pseudo-Boolean Proof Logging for Example CDCL Conflict Analyses



Pseudo-Boolean Proof Logging for Example CDCL Conflict Analyses



pol 7 5 + 2 d 4 + 2 d ;

pol 8 6 + 2 d ;

pol 9 1 + 2 d 10 + 2 d 11 + 2 d ;

$\rightsquigarrow$ Constraint 10 $\doteq u + x \geq 1$

$\rightsquigarrow$ Constraint 11 $\doteq \bar{x} \geq 1$

$\rightsquigarrow$ Constraint 12 $\doteq 0 \geq 1$ ⚡

RUP Revisited

Can define (reverse) unit propagation in a pseudo-Boolean setting [EGMN20]

Constraint C propagates variable x if setting x to “wrong value” would make C unsatisfiable

E.g., if x_5 is false,

$$x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$$

propagates $\bar{x}_4$ (since other coefficients do not add up to 7)

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Risk for confusion!

- Constraint programming people might call this (reverse) integer bounds consistency
 - Does the same thing if we're working with clauses
 - More interesting for general pseudo-Boolean constraints
- SAT people beware: constraints can propagate multiple times and multiple variables

Propagation, Conflict, and Slack

Slack measures how far assignment ρ is from falsifying $\sum_i a_i \ell_i \geq A$

Assuming normalised form:

$$\text{slack}(\sum_i a_i \ell_i \geq A; \rho) = \sum_{i: \rho(\ell_i) \neq 0} a_i - A$$

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ρ	$\text{slack}(C; \rho)$	comment

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$\{\}$	8	

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$\{\}$	8	
$\{\bar{x}_5\}$	3	propagates $\bar{x}_4$ (coefficient > slack)

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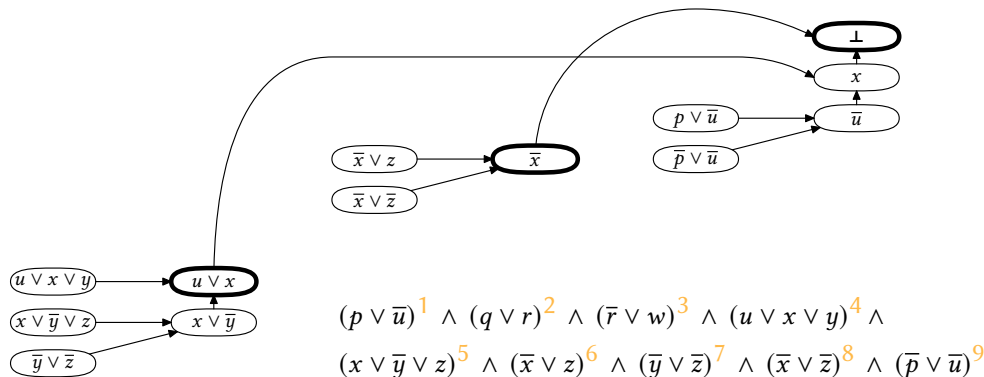
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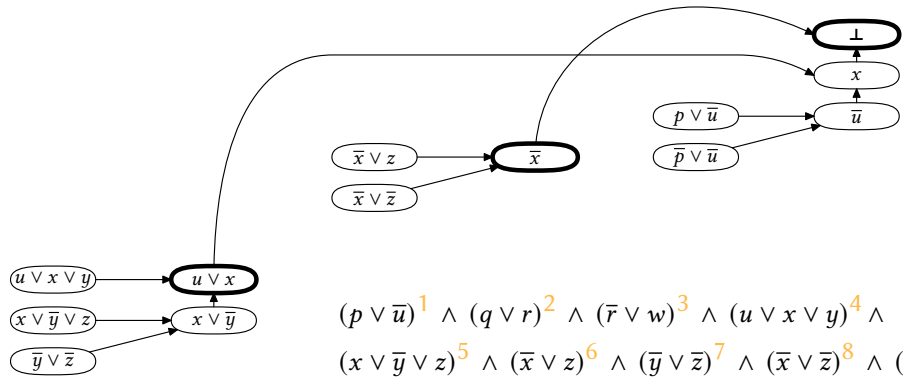
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Note: constraint can be conflicting though not all variables assigned

Pseudo-Boolean Proof Logging for Example CDCL Execution with RUP



Pseudo-Boolean Proof Logging for Example CDCL Execution with RUP



$$(p \vee \bar{u})^1 \wedge (q \vee r)^2 \wedge (\bar{r} \vee w)^3 \wedge (u \vee x \vee y)^4 \wedge (x \vee \bar{y} \vee z)^5 \wedge (\bar{x} \vee z)^6 \wedge (\bar{y} \vee \bar{z})^7 \wedge (\bar{x} \vee \bar{z})^8 \wedge (\bar{p} \vee \bar{u})^9$$

```

rup 1 u 1 x >= 1 ;
rup 1 ~x >= 1 ;
rup >= 1 ;
    
```

```

⇒ Constraint 10 ≐ u + x ≥ 1
⇒ Constraint 11 ≐ x̄ ≥ 1
⇒ Constraint 12 ≐ 0 ≥ 1 ⚡
    
```

Extension Variables, Part 2

Suppose we want new, fresh **reified variable** r encoding

$$r \Leftrightarrow (3x + 2y + z + w \geq 3)$$

This time, introduce constraints

$$3\bar{r} + 3x + 2y + z + w \geq 3 \quad 5r + 3\bar{x} + 2\bar{y} + \bar{z} + \bar{w} \geq 5$$

Again, needs support from the proof system in the form of **strengthening rules**

Proof Logs for “Cutting Planes with Strengthening”

For satisfiable instances: just specify a satisfying assignment.

For unsatisfiability: a sequence of **pseudo-Boolean constraints** in (slight extension of) **OPB format** [RM16]

- Each constraint follows “obviously” from what is known so far
- Either implicitly, by RUP...
- Or by an explicit cutting planes derivation...
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(*) Not actually implemented this way — more details in a few slides ...

Optimisation and Enumeration Problems

Optimisation

- Define an objective $f = \sum_i w_i \ell_i$, $w_i \in \mathbb{Z}$, to minimise subject to the constraints in the formula
- To maximise, negate objective
- Log solution α with `sol1` rule $\Rightarrow$ objective function value $UB = \sum_i w_i \alpha(\ell_i)$
- Add **objective-improving constraint** $\sum_i w_i \ell_i \leq UB - 1$
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Enumeration

- When a solution is found, can log it with solx rule
- Introduces a new constraint saying “not this solution”
- So the proof semantics is “infeasible, except for all the solutions I told you about”

Deleting Constraints

In practice, important to erase constraints to save memory and time during verification

Unsatisfiability proofs: fairly straightforward to deal with from point of view of proof logging

Optimisation and enumeration proofs: significantly more delicate

We will mostly ignore deletions during this tutorial for simplicity and clarity

Pseudo-Boolean Proof Logging — How and Why?

If problem is (special case of) 0-1 integer linear program

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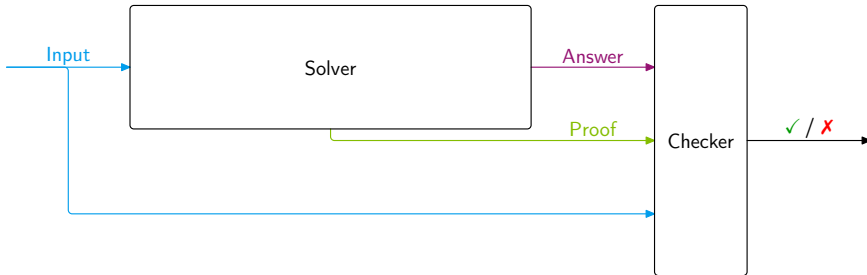
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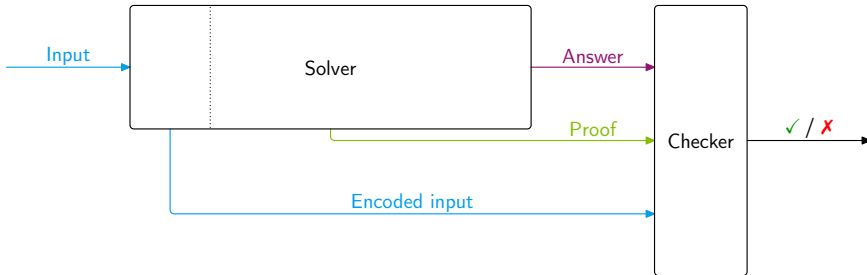
$$7\bar{r} + x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$$

$$9r + \bar{x}_1 + 2x_2 + 3\bar{x}_3 + 4x_4 + 5\bar{x}_5 \geq 9$$

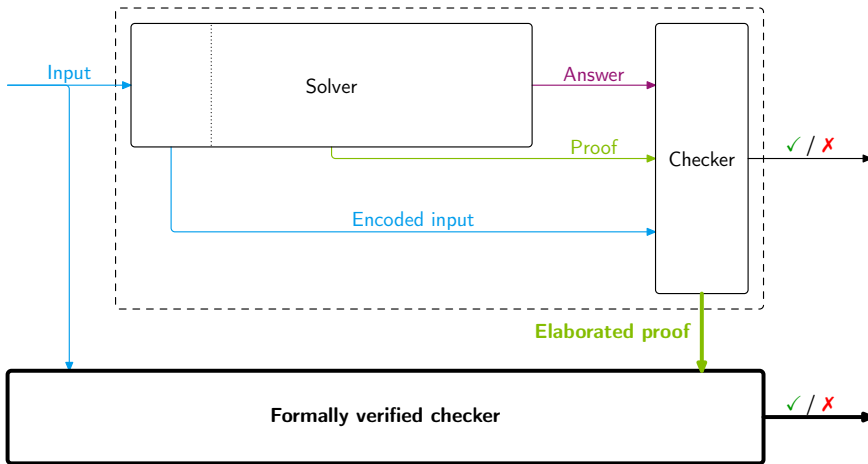
Proof Logging with Formally Verified Checking: Full Workflow



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Strengthening Rules (And the Truth About Extension Variables)

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Wish to allow **without-loss-of-generality** arguments that can derive non-implied constraints

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C is “**redundant**” with respect to F if F and $F \cup \{C\}$ are **equisatisfiable**

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Want to derive

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VERIPB code:

```
red 3 ~r 3 x 2 y 1 z 1 w >= 3 : r -> 0 ;
red 5 r 3 ~x 2 ~y 1 ~z 1 ~w >= 5 : r -> 1 ;
```

Redundance and Dominance Rules for Optimisation

Redundance-based strengthening, optimisation version

Add constraint C to formula F if exists witness substitution ω s.t.

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- Applying ω should **strictly decrease** f
- If so, don't need to show that $C \upharpoonright_{\omega}$ holds!

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- 8 Can't go on forever, so finally reach α' satisfying both F and C

Strength of Dominance Rule

Dominance-based strengthening (stronger, still simplified)

If $C_1, C_2, \dots, C_{m-1}$ have been derived from F (maybe using dominance), then can derive C_m if exists witness substitution ω s.t.

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Only consider F — no need to show that any $C_i \upharpoonright_{\omega}$ implied!

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Further extensions:

- Define dominance rule w.r.t. order independent of objective
- Switch between different orders in same proof
- See [BGMN23, ABB⁺26b] fore more details

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  subproofs for proof goals
qed;
```

```
dom ⟨Constraint C⟩ : ⟨var1⟩ -> ⟨val1⟩ . . . ⟨varN⟩ -> ⟨valN⟩ : subproof
  subproofs for proof goals
qed;
```

- Witness ω should be explicitly specified in proof log
- Subproofs of proof goals should also be explicit
- Except can be skipped for proof goals that “obviously” follow, e.g.,
 - by reverse unit propagation (RUP)
 - by simple syntactic implication from other constraint (as in our reification example)
 - if the proof goal is not affected by the witness substitution ω

Successful Applications of Pseudo-Boolean Proof Logging with VERIPB

Surprisingly, pseudo-Boolean reasoning with strengthening rules sufficient to efficiently certify wide range of combinatorial solving techniques and paradigms:

Successful Applications of Pseudo-Boolean Proof Logging with VERIPB

Surprisingly, pseudo-Boolean reasoning with strengthening rules sufficient to efficiently certify wide range of combinatorial solving techniques and paradigms:

- 1 **Boolean satisfiability (SAT) solving** including advanced techniques such as
 - Gaussian elimination [GN21]
 - symmetry breaking [BGMN23, ABB⁺26b]
- 2 **SAT-based optimization (MaxSAT)** [VDB22, BBN⁺23, BBN⁺24, IOT⁺24, VCB26]
- 3 **Pseudo-Boolean solving and optimization** [GMNO22, KLM⁺25]
- 4 **Graph solving** [GMN20, GMM⁺20, GMM⁺24, DKK⁺26]
- 5 **Dynamic programming and decision diagrams** [DMM⁺24]
- 6 **Presolving in 0–1 integer linear programming** [HOGN24]
- 7 **Constraint programming** [EGMN20, GMN22, MM23, MMN24, MM25]
- 8 **Automated planning** [DHN⁺25]
- 9 **Enumeration of solutions/models** [MNOT26]

VERIPB Resources

VERIPB [tutorials](#), [technical documentation](#), and [more information](#) at
<https://VeriPB.org>



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Lots of [concrete example files](#) at <https://gitlab.com/MIAOresearch/software/VeriPB>

Summing up the First Lecture



- Combinatorial solving and optimisation is a true success story
- But ensuring correctness is a crucial, and not yet satisfactorily addressed, concern
- Certifying solvers producing machine-verifiable proofs of correctness seems like most promising approach
- Cutting planes reasoning with pseudo-Boolean constraints seems to hit a sweet spot between simplicity and expressivity

Summing up the First Lecture



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- Up next:
 - Break
 - And then more details and practical examples

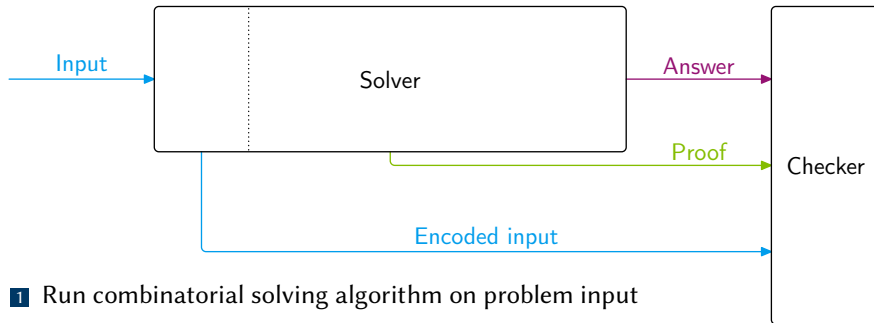
Summing up the First Lecture



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Questions so far?

Recap of Proof Logging Workflow (1/2)



- 1 Run combinatorial solving algorithm on problem input
- 2 Get as output not only answer but also proof
- 3 Feed answer + proof to proof checker together with 0-1 ILP encoding of input

Cutting Planes Rules [CCT87]

Input/model axioms

Literal axioms

Addition

Multiplication for any $c \in \mathbb{N}^+$

Division for any $c \in \mathbb{N}^+$

Saturation for any $c \in \mathbb{N}^+$
(assumes normalised form)

From the input

$$\overline{\ell_i \geq 0}$$

$$\frac{\sum_i a_i \ell_i \geq A \quad \sum_i b_i \ell_i \geq B}{\sum_i (a_i + b_i) \ell_i \geq A + B}$$

$$\frac{\sum_i a_i \ell_i \geq A}{\sum_i c a_i \ell_i \geq cA}$$

$$\frac{\sum_i a_i \ell_i \geq A}{\sum_i \left\lceil \frac{a_i}{c} \right\rceil \ell_i \geq \left\lceil \frac{A}{c} \right\rceil}$$

$$\frac{\sum_i a_i \ell_i \geq A}{\sum_i \min(a_i, A) \cdot \ell_i \geq A}$$

Strengthening Rules in Their (Almost) Full Formal Glory

Witness ω : substitution mapping variables to truth values or literals

Redundance-based strengthening (witness ω show how to “patch assignment”)

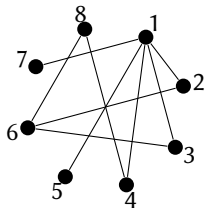
Derive constraint C from $\mathcal{C} \cup \mathcal{D}$ if exists witness ω such that

$$\mathcal{C} \cup \mathcal{D} \cup \{-C\} \vdash (\mathcal{C} \cup \mathcal{D} \cup \{C\}) \upharpoonright_{\omega} \cup \{f \upharpoonright_{\omega} \leq f\} \cup \mathcal{O}(\vec{z} \upharpoonright_{\omega}, \vec{z})$$

Dominance-based strengthening (witness ω “drives down potential”)

Derive constraint C from $\mathcal{C} \cup \mathcal{D}$ if exists witness ω such that

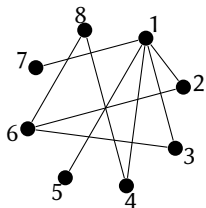
$$\mathcal{C} \cup \mathcal{D} \cup \{-C\} \vdash \mathcal{C}|_{\omega} \cup \{f|_{\omega} \leq f\} \cup \mathcal{O}(\vec{z}|_{\omega}, \vec{z}) \cup \neg \mathcal{O}(\vec{z}, \vec{z}|_{\omega})$$



• • •

More Advanced Example of Redundance-Based Strengthening

Is the following graph 2-colourable (bipartite)?



$$b_1 + b_2 \geq 1$$

$$\bar{b}_1 + \bar{b}_2 \geq 1$$

$$b_1 + b_3 \geq 1$$

$$\bar{b}_1 + \bar{b}_3 \geq 1$$

...

Without loss of generality, node 1 is blue (otherwise we can swap the two colours)

- Improves practical performance
- Affects power of strengthening rules

- Unsatisfiable formulas can turn satisfiable
- Satisfiable formulas can turn unsatisfiable(!)

Enumeration: deletion can reintroduce already enumerated solutions

Solution: distinguish between deletion from core set $\mathcal{C}$ and derived set $\mathcal{D}$

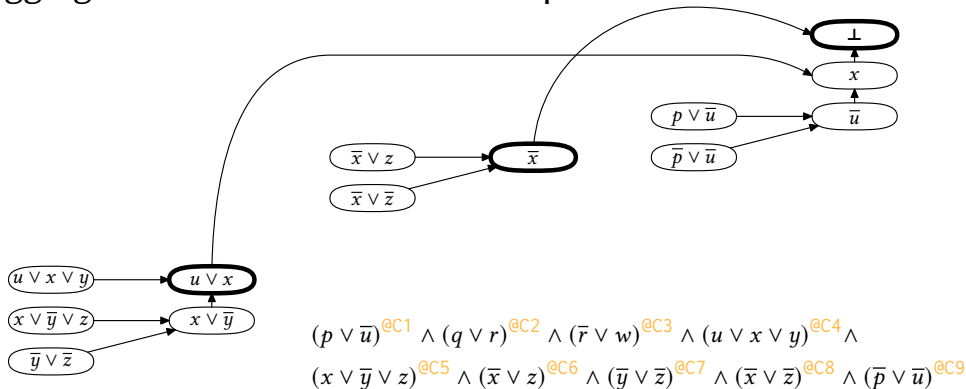
64/128



Constraint identifiers are consecutive numbers 1, 2, 3, ...

- ACP Summer School '26 69/128

Proof Logging with Labels for CDCL Example



```
@C10 pol @C7 @C5 + 2 d @C4 + 2 d ;
```

$$\rightsquigarrow u + x \geq 1$$

```
@C11 pol @C8 @C6 + 2 d ;
```

$$\rightsquigarrow \bar{x} \geq 1$$

@C12 pol @C9 @C1 + 2 d @C10 + 2 d @C11 + 2 d ;

$$\rightsquigarrow 0 \geq 1 \quad \text{⚡}$$

Conversion to Normalised Form: Example

Normalised form is used for convenience and without loss of generality

$$-x_1 + 2x_2 - 3x_3 + 4x_4 - 5x_5 < 0$$

- 1** Make inequality non-strict

$$-x_1 + 2x_2 - 3x_3 + 4x_4 - 5x_5 \leq -1$$

- 2** Multiply by -1 to get greater-than-or-equal

$$x_1 - 2x_2 + 3x_3 - 4x_4 + 5x_5 \geq 1$$

- 3 Replace $-\ell$ by $-(1 - \bar{\ell})$ [where we define $\bar{\bar{x}} \doteq x$]

$$x_1 - 2(1 - \bar{x}_2) + 3x_3 - 4(1 - \bar{x}_4) + 5x_5 \geq 1$$

$$x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$$

For $C \doteq x_1 + 2x_2 + 3x_3 + 4x_4 \geq 5$ we have

- ## Syntactic sugar

$$r_1 \sim r_2 \quad r_3 \Rightarrow 1 \quad x_1 \quad 2 \quad x_2 \quad 3 \quad x_3 \quad 4 \quad x_4 \geq 5 ;$$

Meaning

$$5\bar{r}_1 + 5r_2 + 5\bar{r}_3 + x_1 + 2x_2 + 3x_3 + 4x_4 \geq 5$$

Calculations with Reified Constraints: Concrete Examples

Still with $C \doteq x_1 + 2x_2 + 3x_3 + 4x_4 \geq 5$, so that $\neg C \doteq \bar{x}_1 + 2\bar{x}_2 + 3\bar{x}_3 + 4\bar{x}_4 \geq 6$
(and recall that $\ell + \bar{\ell} = 1$, so that opposing literals cancel and leave 1)

$$\begin{aligned} C + \neg C &\doteq (x_1 + \bar{x}_1) + 2(x_2 + \bar{x}_2) + 3(x_3 + \bar{x}_3) + 4(x_4 + \bar{x}_4) \geq 11 \\ &\doteq 0 \geq 11 - 10 \\ &\doteq 0 \geq 1 \end{aligned}$$

$$\begin{aligned} \text{deg}(C) \cdot (r \geq 1) + (r \Rightarrow C) &\doteq 5(r + \bar{r}) + x_1 + 2x_2 + 3x_3 + 4x_4 \geq 10 \\ &\doteq x_1 + 2x_2 + 3x_3 + 4x_4 \geq 10 - 5 \\ &\doteq C \end{aligned}$$

$$\begin{aligned} C + (r \Leftarrow C) &\doteq 6r + (x_1 + \bar{x}_1) + 2(x_2 + \bar{x}_2) + 3(x_3 + \bar{x}_3) + 4(x_4 + \bar{x}_4) \geq 11 \\ &\doteq 6r \geq 11 - 10 \\ &\doteq \text{deg}(\neg C) \cdot r \geq 1 \end{aligned}$$

Cutting Planes Derivations Under Assumptions

Suppose we have the constraints

$$x_1 + x_2 \geq 1 \quad \rightsquigarrow \quad x_1 + x_2 \geq 1$$

$$x_1 + x_3 \geq 1 \quad \rightsquigarrow \quad x_1 + x_3 \geq 1$$

$$x_2 + x_3 \geq 1 \quad \rightsquigarrow \quad x_2 + x_3 \geq 1$$

$$\cancel{\bar{x}_2} + \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 \quad \rightsquigarrow \quad \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2$$

Assume $r_1 = 0$ and $r_2 = 1$ and simplify

The Same Derivation Without Assumptions

Now forget about the assumptions and run exactly the same derivation on the original constraints:

$$\begin{array}{rcl}
 \text{Add} & \frac{r_1 + x_1 + x_2 \geq 1}{2r_1 + 2x_1 + x_2 + x_3 \geq 2} & \frac{r_1 + x_1 + x_3 \geq 1}{x_2 + x_3 \geq 1} \\
 \text{Add} & \frac{2r_1 + 2x_1 + x_2 + x_3 \geq 2}{2r_1 + 2x_1 + 2x_2 + 2x_3 \geq 3} & \\
 \text{Divide by 2} & \frac{2r_1 + 2x_1 + 2x_2 + 2x_3 \geq 3}{r_1 + x_1 + x_2 + x_3 \geq 2} & \\
 \text{Add} & \frac{r_1 + x_1 + x_2 + x_3 \geq 2}{r_1 + \bar{r}_2 \geq 1} & \bar{r}_2 + \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2
 \end{array}$$

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Now forget about the assumptions and run exactly the same derivation on the original constraints:

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 \text{Divide by 2} & \frac{2r_1 + 2x_1 + 2x_2 + 2x_3 \geq 3}{r_1 + x_1 + x_2 + x_3 \geq 2} & \bar{r}_2 + \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 \\
 \text{Add} & \frac{r_1 + x_1 + x_2 + x_3 \geq 2}{r_1 + \bar{r}_2 \geq 1} &
 \end{array}$$

This yields

$$r_1 + \bar{r}_2 \geq 1 \quad [\text{VERIPB syntax: } 1 \ r1 \ 1 \ \sim r2 \ >= \ 1 \ ;]$$

or equivalently

$$\bar{r}_1 \wedge r_2 \Rightarrow (0 \geq 1) \quad [\text{VERIPB syntax: } \sim r1 \ r2 ==> >= 1 ;]$$

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$$\begin{array}{rcl}
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 \text{Add} & \frac{r_1 + x_1 + x_2 + x_3 \geq 2}{r_1 + \bar{r}_2 \geq 1} & \bar{r}_2 + \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2
 \end{array}$$

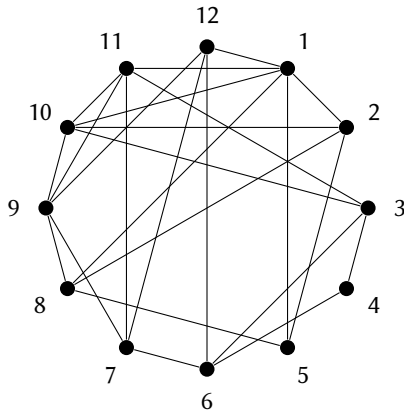
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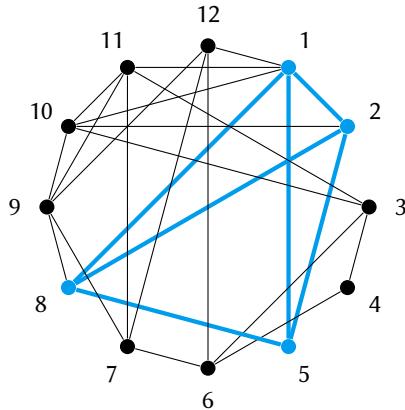
or equivalently

$$\bar{r}_1 \wedge r_2 \Rightarrow (0 \geq 1) \quad [\text{VERIPB syntax: } \sim r1 \ r2 ==> >= 1 ;]$$

That is, our assumptions imply contradiction



The Maximum Clique Problem



There are a lot of dedicated solvers for clique problems

But there are issues:

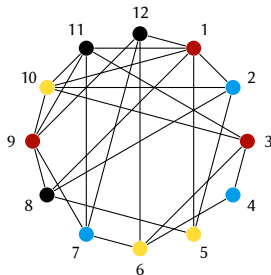
- “State-of-the-art” solvers have been buggy.
- Often undetected: error rate of around 0.1% [MPP19]

Often used inside other solvers

- An off-by-one result can cause much larger errors

Recursive maximum clique algorithm:

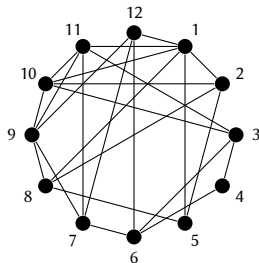
- Pick a vertex v
- Either v is in the clique...
 - Throw away every vertex not adjacent to v
 - If vertices remain, recurse
- ...or v is not in the clique
 - Throw away v (and all incident edges) and pick another vertex



Given a k -colouring of a subgraph, that subgraph cannot have a clique of more than k vertices

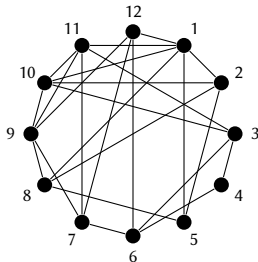
We can use $|C| + \#colours(P)$ as a bound, for any colouring

A Pseudo-Boolean Encoding for Clique (in OPB Format)



```
* #variable= 12 #constraint= 41
min: 1 ~x1 1 ~x2 1 ~x3 1 ~x4 . . . and so on . . . 1 ~x11 1 ~x12 ;
@noedge1_3 -1 x3 -1 x1 >= -1 ;
@noedge2_3 -1 x3 -1 x2 >= -1 ;
@noedge1_4 -1 x4 -1 x1 >= -1 ;
* . . . and a further 38 similar lines for the remaining non-edges
```

A Pseudo-Boolean Encoding for Clique (in OPB Format)



```
* #variable= 12 #constraint= 41
```

```
min: 1 ~x1 1 ~x2 1 ~x3 1 ~x4 . . . and so on . . . 1 ~x11 1 ~x12 ;
```

```
@noedge1_3 -1 x3 -1 x1 >= -1 ;
```

```
@noedge2_3 -1 x3 -1 x2 >= -1 ;
```

```
@noedge1_4 -1 x4 -1 x1 >= -1 ;
```

```
* . . . and a further 38 similar lines for the remaining non-edges
```

$x_i = 1$ means “vertex i is in the clique”

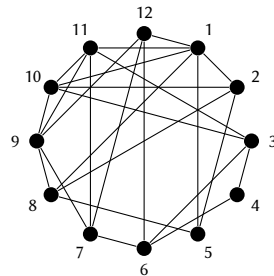
VERIPB proof minimises number of vertices **not** selected, so 4-clique yields objective value $12 - 4 = 8$

First Attempt at a Proof

```

pseudo-Boolean proof version 3.0
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;

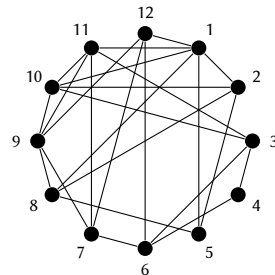
```



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pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```



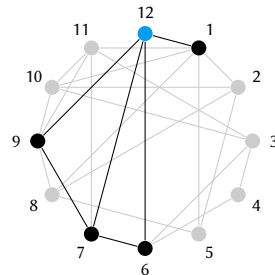
Start with a header

The problem constraints are loaded for us

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

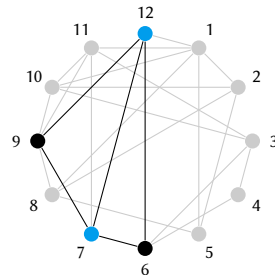


Branch accepting 12
Throw away non-adjacent vertices

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```



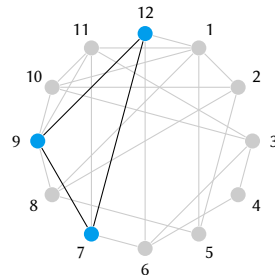
Branch also accepting 7
Throw away non-adjacent vertices

First Attempt at a Proof

```

pseudo-Boolean proof version 3.0
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;

```

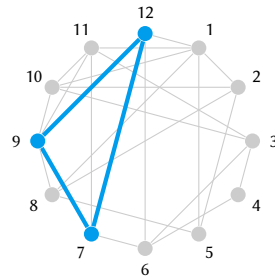


Branch also accepting 9
Throw away non-adjacent vertices

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;  
rup 1 ~x12 1 ~x7 >= 1 ;  
rup 1 ~x12 >= 1 ;  
rup 1 ~x11 1 ~x10 >= 1 ;  
rup 1 ~x11 >= 1 ;  
@obj soli x1 x2 x5 x8 ;  
rup 1 ~x8 1 ~x5 >= 1 ;  
rup 1 ~x8 >= 1 ;  
rup >= 1 ;  
output NONE ;  
conclusion BOUNDS 8 8 ;  
end pseudo-Boolean proof ;
```



We branched on 12, 7, 9
Found a new incumbent
Now looking for a ≥ 4 vertex clique
“@obj” names the constraint this creates

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
```

```
rup 1 ~x12 1 ~x7 >= 1 ;
```

```
rup 1 ~x12 >= 1 ;
```

```
rup 1 ~x11 1 ~x10 >= 1 ;
```

```
rup 1 ~x11 >= 1 ;
```

```
@obj soli x1 x2 x5 x8 ;
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```
rup 1 ~x8 1 ~x5 >= 1 ;
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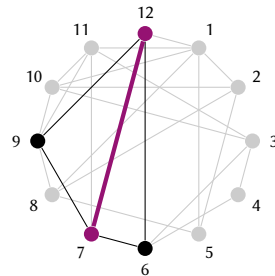
```
rup 1 ~x8 >= 1 ;
```

```
rup >= 1 ;
```

```
output NONE ;
```

```
conclusion BOUNDS 8 8 ;
```

```
end pseudo-Boolean proof ;
```

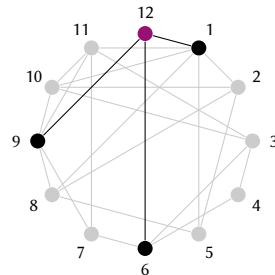


Backtrack from 12, 7
 9 explored already, only 6 feasible
 No ≥ 4 vertex clique possible
 Effectively this deletes the 7–12 edge

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

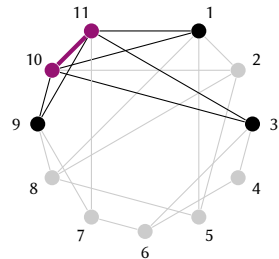


Backtrack from 12
 Only 1, 6 and 9 feasible (1-colourable)
 No ≥ 4 vertex clique possible
 Effectively this deletes vertex 12

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

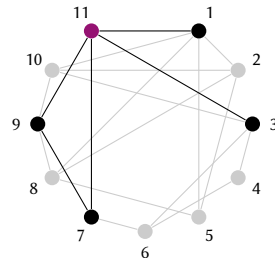


Branch on 11 then 10
Only 1, 3 and 9 feasible (1-colourable)
No ≥ 4 vertex clique possible
Backtrack, deleting the edge

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

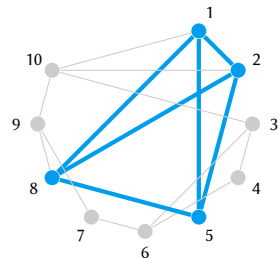


Backtrack from 11
2-colourable, so no ≥ 4 clique
Delete the vertex

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```



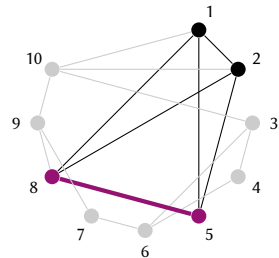
Branch on 8, 5, 1, 2
Find a new incumbent
Now looking for a ≥ 5 vertex clique

First Attempt at a Proof

```

pseudo-Boolean proof version 3.0
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;

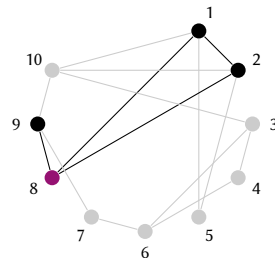
```



Backtrack from 8, 5
Only 4 vertices; can't have a ≥ 5 clique
Delete the edge

First Attempt at a Proof

```
pseudo-Boolean proof version 3.0
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

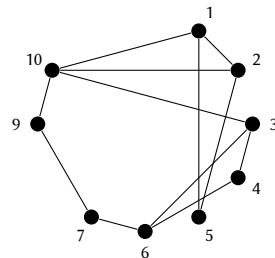


Backtrack from 8
Still not enough vertices
Delete the vertex

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```



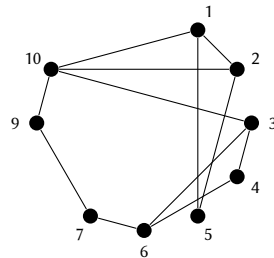
Remaining graph is 3-colourable
Backtrack from root node

First Attempt at a Proof

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
rup 1 ~x12 >= 1 ;
rup 1 ~x11 1 ~x10 >= 1 ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
rup 1 ~x8 >= 1 ;
rup >= 1 ;
```

```
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```



Finish with what we've concluded
 We specify a lower and an upper bound
 Remember we're minimising the number of vertices we *don't* select, so a 4-clique has an objective value of $12 - 4 = 8$
 ("output" is for features we're not using)

Verifying This Proof (Or Not...)

```
$ veripb clique.opb clique-attempt-one.pbp
Running VeriPB version 3.0.2
Error: Checking error at clique-attempt-one.pbp:5
```

Caused by:

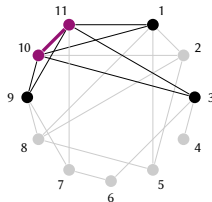
```
The constraint is not implied by reverse unit propagation
(RUP) from core and derived database. Rerun with the option
'--trace-failed' to see the propagations. . . .
```

Verifying This Proof (Or Not...)

```
$ veripb clique.opb clique-attempt-one.pbp
Running VeriPB version 3.0.2
Error: Checking error at clique-attempt-one.pbp:5
```

Caused by:

The constraint is not implied by reverse unit propagation (RUP) from core and derived database. Rerun with the option '--trace-failed' to see the propagations. . . .



Verifying This Proof (Or Not...)

```
$ veripb --trace clique.opb clique-attempt-one.pbp
Running VeriPB version 3.0.2
  Objective: min 1 ~x1 1 ~x2 . . . 1 ~x11 1 ~x12 0
  ConstraintId 1: 1 ~x1 1 ~x3 >= 1
  ConstraintId 2: 1 ~x2 1 ~x3 >= 1
  . . .
  ConstraintId 41: 1 ~x11 1 ~x12 >= 1
line   1: pseudo-Boolean proof version 3.0
line   2: @obj soli x7 x9 x12 ;
  ConstraintId 42: 1 x1 1 x2 . . . 1 x11 1 x12 >= 4
line   3: rup 1 ~x12 1 ~x7 >= 1 ;
  ConstraintId 43: 1 ~x7 1 ~x12 >= 1
line   4: rup 1 ~x12 >= 1 ;
  ConstraintId 44: 1 ~x12 >= 1
line   5: rup 1 ~x11 1 ~x10 >= 1 ;
Error: Checking error at clique-attempt-one.pbp:5
```


Verifying This Proof (Or Not...)

```
$ veripb --trace-failed clique.opb clique-attempt-one.pbp
```

```
Running VeriPB version 3.0.2
```

```
Propagation check failed! The propagation had the following trail:
```

```
  propagations in format: <assignment> (<reason constraint>)
```

```
    ~x12 (1 ~x12 >= 1)
```

```
    x10 (1 x10 1 x11 >= 2)
```

```
    x11 (1 x10 1 x11 >= 2)
```

```
    ~x4 (1 ~x4 1 ~x10 >= 1)
```

```
    ~x5 (1 ~x5 1 ~x10 >= 1)
```

```
    ~x6 (1 ~x6 1 ~x10 >= 1)
```

```
    ~x7 (1 ~x7 1 ~x10 >= 1)
```

```
    ~x8 (1 ~x8 1 ~x10 >= 1)
```

```
    ~x2 (1 ~x2 1 ~x11 >= 1)
```

```
Error: Checking error at clique-attempt-one.pbp:5
```

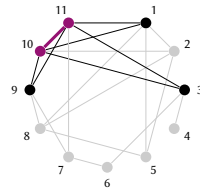
The Problem of Counting

The proof checker isn't smart enough to work out that

- vertices 1, 3 and 9 can all be given the same colour...
- and each colour class contributes at most one vertex to a clique...
- so “find me a clique with more than three vertices” can't be satisfied once we've accepted both 10 and 11

Unit propagation never sees this:

- The bound needs **counting**
- But the encoding only ever talks about **pairs** of vertices



Dealing With Colourings

So the colour bound doesn't follow by reverse unit propagation...

But we can lazily recover at-most-one constraints for each colour class!

Dealing With Colourings

So the colour bound doesn't follow by reverse unit propagation...

But we can lazily recover at-most-one constraints for each colour class!

$$\begin{aligned}
 &(\bar{x}_1 + \bar{x}_6 \geq 1) \\
 + &(\bar{x}_1 + \bar{x}_9 \geq 1) & = 2\bar{x}_1 + \bar{x}_6 + \bar{x}_9 \geq 2 \\
 + &(\bar{x}_6 + \bar{x}_9 \geq 1) & = 2\bar{x}_1 + 2\bar{x}_6 + 2\bar{x}_9 \geq 3 \\
 & & / 2 & = \bar{x}_1 + \bar{x}_6 + \bar{x}_9 \geq 2 \\
 & & & \text{i.e. } x_1 + x_6 + x_9 \leq 1
 \end{aligned}$$

Dealing With Colourings

So the colour bound doesn't follow by reverse unit propagation...

But we can lazily recover at-most-one constraints for each colour class!

$$\begin{aligned}
 &(\bar{x}_1 + \bar{x}_6 \geq 1) \\
 + &(\bar{x}_1 + \bar{x}_9 \geq 1) &&= 2\bar{x}_1 + \bar{x}_6 + \bar{x}_9 \geq 2 \\
 + &(\bar{x}_6 + \bar{x}_9 \geq 1) &&= 2\bar{x}_1 + 2\bar{x}_6 + 2\bar{x}_9 \geq 3 \\
 &&&/ 2 &&= \bar{x}_1 + \bar{x}_6 + \bar{x}_9 \geq 2 \\
 &&&&&\text{i.e. } x_1 + x_6 + x_9 \leq 1
 \end{aligned}$$

This generalises to colour classes of any size v

- Each non-edge is used exactly once, $v(v-1)$ additions
- $v-3$ multiplications and $v-2$ divisions

Solvers don't need to "understand" cutting planes to write out this derivation

Details on At-Most-One Constraint Recovery (3 Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

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- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\text{Add} \quad \frac{\bar{x}_1 + \bar{x}_2 \geq 1 \quad \bar{x}_1 + \bar{x}_3 \geq 1}{2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2}$$

Details on At-Most-One Constraint Recovery (3 Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
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- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 & \bar{x}_1 + \bar{x}_2 \geq 1 & \bar{x}_1 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 & \bar{x}_2 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 3
 \end{array}$$

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- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 & \bar{x}_1 + \bar{x}_2 \geq 1 & \bar{x}_1 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 & \bar{x}_2 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 3 \\
 \text{Divide by 2} & \hline
 & \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq \frac{3}{2}
 \end{array}$$

Details on At-Most-One Constraint Recovery (3 Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 & \bar{x}_1 + \bar{x}_2 \geq 1 & \bar{x}_1 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 & \bar{x}_2 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 3 \\
 \text{Divide by 2} & \hline
 & \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq \left\lceil \frac{3}{2} \right\rceil
 \end{array}$$

Details on At-Most-One Constraint Recovery (3 Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 & \bar{x}_1 + \bar{x}_2 \geq 1 & \bar{x}_1 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 & \bar{x}_2 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 3 \\
 \text{Divide by 2} & \hline
 & \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2
 \end{array}$$

Details on At-Most-One Constraint Recovery (3 Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 & \bar{x}_1 + \bar{x}_2 \geq 1 & \bar{x}_1 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2 & \bar{x}_2 + \bar{x}_3 \geq 1 \\
 \text{Add} & \hline
 & 2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 3 \\
 \text{Divide by 2} & \hline
 & \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2
 \end{array}$$

Assuming constraint $\bar{x}_i + \bar{x}_j \geq 1$ has label @C*ij*, get VERiPB code

```
pol @C12 @C13 + @C23 + 2 d ;
```

Details on At-Most-One Constraint Recovery (4 and More Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2$$

Details on At-Most-One Constraint Recovery (4 and More Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\text{Multiply by 2} \quad \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2}{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4}$$

Details on At-Most-One Constraint Recovery (4 and More Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 \text{Multiply by 2} & \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2}{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4} & \bar{x}_1 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4}{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5} & \bar{x}_2 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5}{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6} & \bar{x}_3 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6}{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7} &
 \end{array}$$

Details on At-Most-One Constraint Recovery (4 and More Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 \text{Multiply by 2} & \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2}{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4} & \bar{x}_1 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4}{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5} & \bar{x}_2 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5}{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6} & \bar{x}_3 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6}{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7} & \\
 \text{Divide by 3} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7}{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \bar{x}_4 \geq \left\lceil \frac{7}{3} \right\rceil} &
 \end{array}$$

Details on At-Most-One Constraint Recovery (4 and More Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
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- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 \text{Multiply by 2} & \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2}{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4} & \bar{x}_1 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4}{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5} & \bar{x}_2 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5}{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6} & \bar{x}_3 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6}{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7} & \\
 \text{Divide by 3} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7}{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \bar{x}_4 \geq 3} &
 \end{array}$$

Details on At-Most-One Constraint Recovery (4 and More Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
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$$\begin{array}{rcl}
 \text{Multiply by 2} & \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2}{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4} & \bar{x}_1 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4}{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5} & \bar{x}_2 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5}{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6} & \bar{x}_3 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6}{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7} & \\
 \text{Divide by 3} & \frac{3\bar{x}_1 + 3\bar{x}_2 + 3\bar{x}_3 + 3\bar{x}_4 \geq 7}{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \bar{x}_4 \geq 3} &
 \end{array}$$

Assuming constraint $\bar{x}_i + \bar{x}_j \geq 1$ has label @C_{ij}, VERIPB code for full derivation is

```
pol @C12 @C13 + @C23 + 2 d 2 * @C14 + @C24 + @C34 + 3 d ;
```

Details on At-Most-One Constraint Recovery (4 and More Variables)

- Derive constraint $x_1 + \dots + x_n \leq 1$ from $\binom{n}{2}$ constraints $x_i + x_j \leq 1$
- I.e., $\bar{x}_1 + \dots + \bar{x}_n \geq n - 1$ from constraints $\bar{x}_i + \bar{x}_j \geq 1$ in normalised form
- Need this to prove that **at most one vertex per colour class** can be a member of the clique

$$\begin{array}{rcl}
 \text{Multiply by 2} & \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 \geq 2}{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4} & \bar{x}_1 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{2\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 \geq 4}{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5} & \bar{x}_2 + \bar{x}_4 \geq 1 \\
 \text{Add} & \frac{3\bar{x}_1 + 2\bar{x}_2 + 2\bar{x}_3 + \bar{x}_4 \geq 5}{3\bar{x}_1 + 3\bar{x}_2 + 2\bar{x}_3 + 2\bar{x}_4 \geq 6} & \bar{x}_3 + \bar{x}_4 \geq 1 \\
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```
pol @C12 @C13 + @C23 + 2 d 2 * @C14 + @C24 + @C34 + 3 d ;
```

(and this pattern generalizes to any number of variables)

What The Revised Proof with Explicit Counting Looks Like

```
pseudo-Boolean proof version 3.0
```

```
@obj soli x7 x9 x12 ;
```

```
rup 1 ~x12 1 ~x7 >= 1 ;
```

```
pol @noedge1_6 @noedge1_9 + @noedge6_9 + 2 d @obj + ;
```

```
rup 1 ~x12 >= 1 ;
```

```
pol @noedge1_3 @noedge1_9 + @noedge3_9 + 2 d @obj + ;
```

```
rup 1 ~x11 1 ~x10 >= 1 ;
```

```
pol @noedge1_3 @noedge1_7 + @noedge3_7 + 2 d @obj + ;
```

```
rup 1 ~x11 >= 1 ;
```

```
@obj soli x1 x2 x5 x8 ;
```

```
rup 1 ~x8 1 ~x5 >= 1 ;
```

```
pol @obj @noedge1_9 + ;
```

```
rup 1 ~x8 >= 1 ;
```

```
pol @noedge1_3 @noedge1_7 + @noedge3_7 + 2 d @noedge2_4 @noedge2_9 + @noedge4_9 + 2 d +  
    @noedge5_6 @noedge5_10 + @noedge6_10 + 2 d + @obj + ;
```

```
rup >= 1 ;
```

```
output NONE ;
```

```
conclusion BOUNDS 8 8 ;
```

```
end pseudo-Boolean proof ;
```

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```

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```
end pseudo-Boolean proof ;
```

Failing RUP steps have become **pol** derivations: add three non-adjacency constraints together, divide by two, then add the result to the solution-improving constraint

The labels come straight out of the OPB file

What The Revised Proof with Explicit Counting Looks Like

pseudo-Boolean proof version 3.0

```
@obj soli x7 x9 x12 ;
rup 1 ~x12 1 ~x7 >= 1 ;
pol @noedge1_6 @noedge1_9 + @noedge6_9 + 2 d @obj + ;
rup 1 ~x12 >= 1 ;
pol @noedge1_3 @noedge1_9 + @noedge3_9 + 2 d @obj + ;
rup 1 ~x11 1 ~x10 >= 1 ;
pol @noedge1_3 @noedge1_7 + @noedge3_7 + 2 d @obj + ;
rup 1 ~x11 >= 1 ;
@obj soli x1 x2 x5 x8 ;
rup 1 ~x8 1 ~x5 >= 1 ;
pol @obj @noedge1_9 + ;
rup 1 ~x8 >= 1 ;
```

```
pol @noedge1_3 @noedge1_7 + @noedge3_7 + 2 d @noedge2_4 @noedge2_9 + @noedge4_9 + 2 d +
  @noedge5_6 @noedge5_10 + @noedge6_10 + 2 d + @obj + ;
```

```
rup >= 1 ;
output NONE ;
conclusion BOUNDS 8 8 ;
end pseudo-Boolean proof ;
```

Three colour classes here, so three at-most-one constraints, all added to the solution-improving constraint

Verifying This Proof (For Real, This Time)

```
$ veripb clique.opb clique-new-attempt.pbp
Running VeriPB version 3.0.2
s VERIFIED BOUNDS 8 <= obj <= 8
```

No errors this time — every step is justified

Note that the checker **knows nothing** about

- cliques,
- colourings, or
- bound functions

Verifying This Proof (For Real, This Time)

```
$ veripb --trace clique.opb clique-new-attempt.pbp
Running VeriPB version 3.0.2
Objective: min 1 ~x1 1 ~x2 . . . 1 ~x11 1 ~x12 0
ConstraintId 1: 1 ~x1 1 ~x3 >= 1
. . .
ConstraintId 41: 1 ~x11 1 ~x12 >= 1
line 2: @obj soli x7 x9 x12 ;
ConstraintId 42: 1 x1 1 x2 . . . 1 x11 1 x12 >= 4
line 3: rup 1 ~x12 1 ~x7 >= 1 ;
ConstraintId 43: 1 ~x7 1 ~x12 >= 1
line 4: pol @noedge1_6 @noedge1_9 + @noedge6_9 + 2 d @obj + ;
ConstraintId 44: 1 x2 1 x3 1 x4 1 x5 1 x7 1 x8 1 x10 1 x11 1 x12 >= 3
. . .
line 16: rup >= 1 ;
ConstraintId 55: >= 1
s VERIFIED BOUNDS 8 <= obj <= 8
```


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Cutting planes to the rescue!

- Turns out to be strong enough for all of these cases
- Proof length scales linearly with solver reasoning time (asymptotically)

Different Clique Algorithms

Different search orders?

- ✓ Irrelevant for proof logging

Using local search to initialise?

- ✓ Just log the incumbent

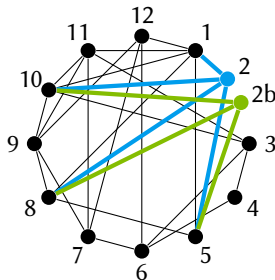
Different bound functions?

- Is cutting planes strong enough to justify every useful bound function ever invented?
- So far, seems like it...

Weighted cliques?

- ✓ Multiply a colour class by its largest weight
- ✓ Also works for vertices “split between colour classes”

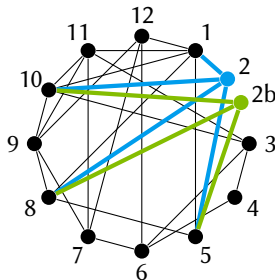
Lazy Global Domination for Maximum Clique [MP16]



Can ignore vertex 2b

- Every neighbour of 2b is also a neighbour of 2
- This is an example of **dominance** reasoning

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Dominance rule can justify this

- Even when detected dynamically during search

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- VERIPB can **elaborate** a proof into a simpler **kernel format**
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- CAKEPB (written in CAKEML and proved correct in HOL4)
 - reads the **original graph file**
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- So the answer is verified in terms of the original input

End-to-End Verification for Maximum Clique

```
$ glasgow_clique_solver brock200_4.clq
recursions = 56613
clique = 17 (12 19 28 29 38 54 65 71 79 93 117 127 139 161 165 186 192)
runtime = 0.02153s

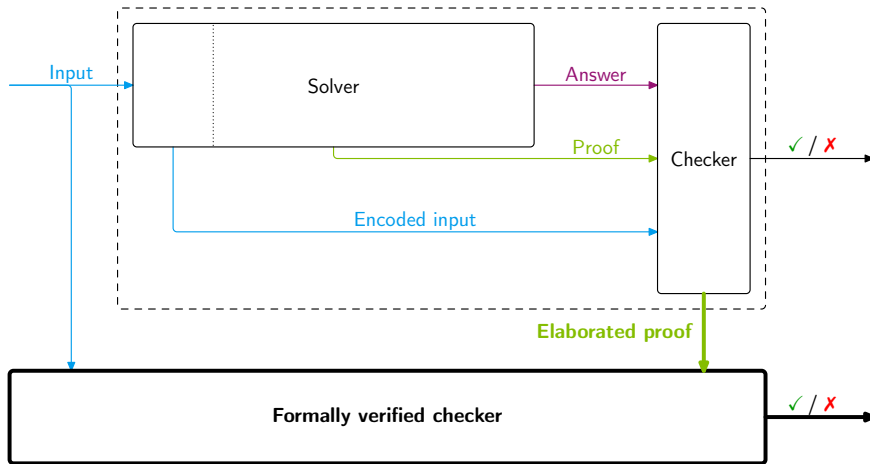
$ glasgow_clique_solver brock200_4.clq --prove brock200_4
runtime = 0.266475s

$ cake_pb_clique brock200_4.clq > brock200_4.verifiedopb

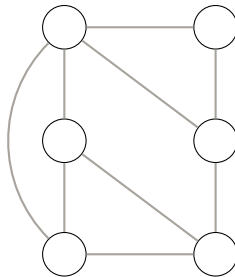
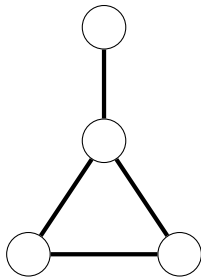
$ veripb --elaborate brock200_4.corepb brock200_4.verifiedopb brock200_4.pbp
s VERIFIED BOUNDS 183 <= obj <= 183

$ cake_pb_clique brock200_4.clq brock200_4.corepb
s VERIFIED MAX CLIQUE SIZE |CLIQUE| = 17
```

Workflow for Solving with End-to-End Certified Results

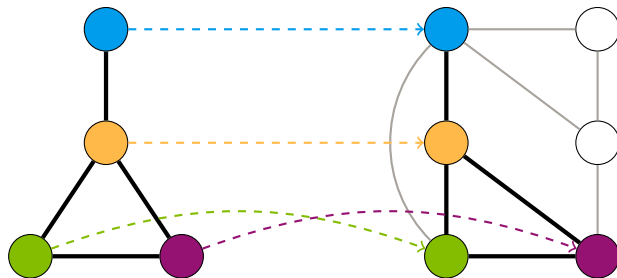


Subgraph Isomorphism



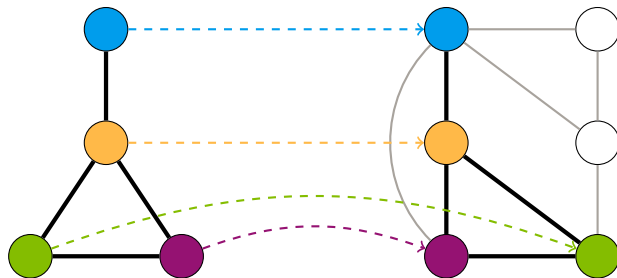
- Find the **pattern** inside the **target**
- Applications in compilers, biochemistry, model checking, pattern recognition, ...
- Often want to find **all** matches

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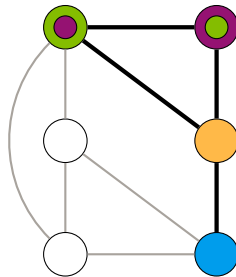
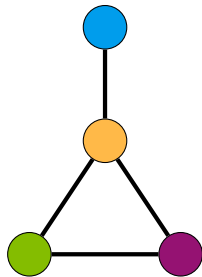
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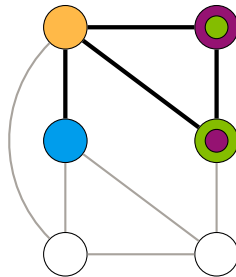
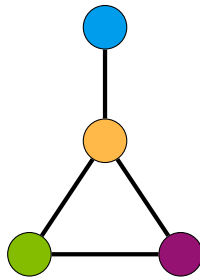
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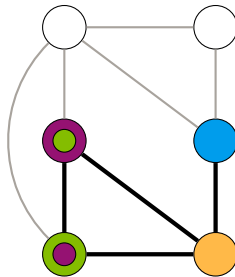
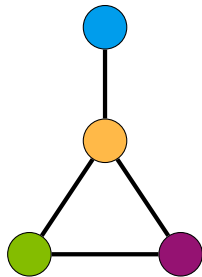
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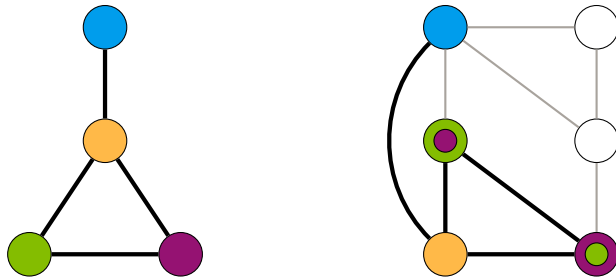
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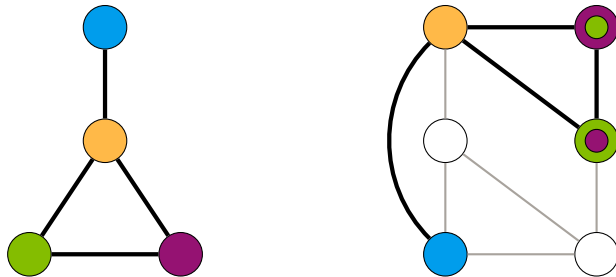
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Subgraph Isomorphism in Pseudo-Boolean Form

Each pattern vertex gets a target vertex:

$$\sum_{t \in V(T)} x_{p,t} = 1 \qquad p \in V(P)$$

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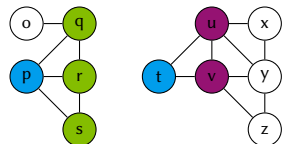
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Adjacency constraints: if p maps to t , then neighbours of p map to neighbours of t :

$$\bar{x}_{p,t} + \sum_{u \in N(t)} x_{q,u} \geq 1 \quad p \in V(P), q \in N(p), t \in V(T)$$

Degree Reasoning in Cutting Planes

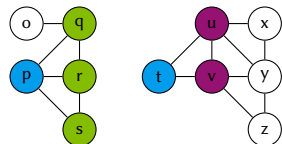


Pattern vertex p of degree $\deg(p)$ can never be mapped to target vertex t of degree $< \deg(p)$

Observe $N(p) = \{q, r, s\}$ and $N(t) = \{u, v\}$

We wish to derive $\bar{x}_{p,t} \geq 1$

Degree Reasoning in Cutting Planes



Adjacency:

$$\bar{x}_{p,t} + x_{q,u} + x_{q,v} \geq 1$$

$$\bar{x}_{p,t} + x_{r,u} + x_{r,v} \geq 1$$

$$\bar{x}_{p,t} + x_{s,u} + x_{s,v} \geq 1$$

Injectivity:

$$-x_{o,u} + -x_{p,u} + -x_{q,u} + -x_{r,u} + -x_{s,u} \geq -1$$

$$-x_{o,v} + -x_{p,v} + -x_{q,v} + -x_{r,v} + -x_{s,v} \geq -1$$

Literal axioms:

$$x_{o,u} \geq 0$$

$$x_{o,v} \geq 0$$

$$x_{p,u} \geq 0$$

$$x_{p,v} \geq 0$$

Add these together ...

$$3 \cdot \bar{x}_{p,t} \geq 1$$

Degree Reasoning in VERIPB

```
pol @adj_p_t_q @adj_p_t_r + @adj_p_t_s + % sum adjacency constraints
    @inj_u + @inj_v + % sum injectivity constraints
    xo_u + xo_v + % cancel stray xo_* >= 0
    xp_u + xp_v + % cancel stray xp_* >= 0
    3 d ; % divide, and we're done
```

The labels come from the OPB file, so solver doesn't need to keep track of constraint IDs

Limitations

Why trust the encoding?

- ✓ For clique and subgraph isomorphism this is now taken care of [GMM⁺24]
- Has to be done again for each new problem, but not too bad

Compiling Table Constraints

Constraints can be specified **extensionally** as list of feasible tuples, called a **table**

Variable assignments must match some row in table

Given table constraint

$$(A, B, C) \in [(1, 2, 3), (1, 3, 4), (2, 2, 5)]$$

define

$$3\bar{t}_1 + a_{=1} + b_{=2} + c_{=3} \geq 3$$

$$\text{i.e., } t_1 \Rightarrow (a_{=1} \wedge b_{=2} \wedge c_{=3})$$

$$3\bar{t}_2 + a_{=1} + b_{=3} + c_{=4} \geq 3$$

$$\text{i.e., } t_2 \Rightarrow (a_{=1} \wedge b_{=3} \wedge c_{=4})$$

$$3\bar{t}_3 + a_{-2} + b_{-2} + c_{-5} \geq 3$$

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using newly introduced tuple selector variables

$$t_1 + t_2 + t_3 = 1$$

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Tuple variables **used for proof logging** — need not be known by solver (except for printing proofs)

on of

No need to justify guesses or decisions — only justify backtracking

Anything the constraint programming solver knows must follow from **unit propagation** of guessed assignments on **constraints in proof log**

Key idea

Anything the constraint programming solver knows must follow from **unit propagation** of guessed assignments on **constraints in proof log**

If it follows from unit propagation on the encoding, nothing needed

Some propagators and encodings need RUP steps for inferences

- A lot of propagators are effectively “doing a little bit of lookahead” but in an efficient way

A few need explicit cutting planes justifications written to the proof log

- **Linear inequalities** just need to multiply and add
- **All-different** needs a bit more

$$Z \in \{1, 3\}$$

$$V \in \{1, 4, 5\}$$

$$W \in \{1, 2, 3\} \quad w_1 + w_2 + w_3 \geq 1 \quad [W \text{ takes some value}]$$

$$X \in \{ \quad 2 \quad 3 \quad \}$$

$$Y \in \{1, 3\}$$

$$Z \in \{1, 3\}$$

$$\begin{array}{llllll} V \in \{1 & & 4 & 5\} & & \\ W \in \{1 & 2 & 3 & & \} & w_{=1} + w_{=2} + w_{=3} \geq 1 \quad [W \text{ takes some value}] \\ X \in \{ & 2 & 3 & & \} & x_{=2} + x_{=3} \geq 1 \quad [X \text{ takes some value}] \\ Y \in \{1 & & 3 & & \} & y_{=1} + y_{=3} \geq 1 \quad [Y \text{ takes some value}] \\ Z \in \{1 & & 3 & & \} & z_{=1} + z_{=3} \geq 1 \quad [Z \text{ takes some value}] \end{array}$$

Justifying All-Different Failures

$$Z \in \{1, 3\} \quad z_{=1} + z_{=3} \geq 1 \quad [Z \text{ takes some value}]$$

$$\rightarrow -w_{=3} + -x_{=3} + -y_{=3} + -z_{=3} \geq -1 \quad [\text{At most one variable} = 3]$$

Justifying All-Different Failures

$$-v_{=1} \geq 1 \quad [\text{Sum all constraints so far}]$$

Justifying All-Different Failures

$V \in \{1 \quad 4 \quad 5\}$				
$W \in \{1 \quad 2 \quad 3 \quad \}$	$w_{=1} +$	$w_{=2} +$	$w_{=3}$	≥ 1 [W takes some value]
$X \in \{ \quad 2 \quad 3 \quad \}$		$x_{=2} +$	$x_{=3}$	≥ 1 [X takes some value]
$Y \in \{1 \quad 3 \quad \}$	$y_{=1}$	$+$	$y_{=3}$	≥ 1 [Y takes some value]
$Z \in \{1 \quad 3 \quad \}$	$z_{=1}$	$+$	$z_{=3}$	≥ 1 [Z takes some value]

$$\begin{array}{llll}
\rightarrow & -v_{=1} + -w_{=1} + & -y_{=1} + -z_{=1} \geq -1 & [\text{At most one variable} = 1] \\
\rightarrow & & -w_{=2} + -x_{=2} & \geq -1 \quad [\text{At most one variable} = 2] \\
\rightarrow & & -w_{=3} + -x_{=3} + -y_{=3} + -z_{=3} \geq -1 & [\text{At most one variable} = 3] \\
\\
& -v_{=1} & & \geq 1 \quad [\text{Sum all constraints so far}] \\
& v_{=1} & & \geq 0 \quad [\text{Variable } v_{=1} \text{ non-negative}]
\end{array}$$

Justifying All-Different Failures

$$Z \in \{1 \quad 3 \quad \} \quad z_{=1} \quad + \quad z_{=3} \quad \geq \quad 1 \quad [Z \text{ takes some value }]$$

$$\rightarrow -w_{=3} + -x_{=3} + -y_{=3} + -z_{=3} \geq -1 \quad [\text{At most one variable} = 3]$$

$$v_{=1} \geq 0 \quad [\text{Variable } v_{=1} \text{ non-negative}]$$

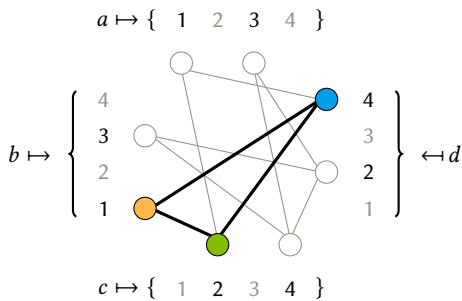
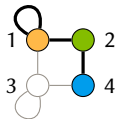
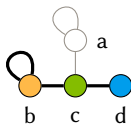
$$0 \leq 1 \quad [\text{Sum above two constraints}]$$

Reformulation

Auto-tabulation is possible

- Heavy use of extension variables

Can re-encode max common subgraph as clique problem without changing PB encoding



How do we know we're giving a proof for the problem the user actually specified?

Need modelling language with formally specified semantics...

The *Glasgow* Constraint Solver

Support for 37 (families of) constraints (and counting), with all inferences certified:

Abs, AllDifferent, AllEqual, Among, AtMostOne, BinPacking, Circuit, Comparison, Count, Cumulative, Diffn, Disjunctive, DivMod, Element, Equals, GCC, In, Increasing, Inverse, Knapsack, Lex, Linear, Logical, MDD, MinDistance, MinMax, Multiply, NValue, Parity, PlusMinus, Power, Regular, SeqPrecedeChain, SmartTable, Sort, Table, ValuePrecede

Also **certifies reformulations** not written in but inferred from input:

- Auto-tabulation: Identify small subproblem, solve it, and replace by table
- Capacity strengthening by integrality, conflict cliques across resources, lifted cover cuts:
Extract information implied by scheduling problem constraints
- Difference logic: Spot that a pile of two-term inequalities is a precedence network, and hand to temporal propagator

Performance of pseudo-Boolean proof logging and checking

- Faster proof logging and checking!
- Trim proofs while checking as in SAT solving [HHW13a] (*upcoming work [ABB⁺26a]*)
- Compress proof files using binary format

- Model counting
- Mixed integer linear programming (*suggested extension of VERIPB in [DEGH23]*)
- Satisfiability modulo theories (SMT) solving (*Lots of work in, e.g., [HS22, BBC⁺23, RSB⁺26, RSS⁺26]*)

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- Use proof logs for algorithm analysis or explainability purposes
- Lots of other challenging problems and interesting ideas
- Talk to us if you want to join the pseudo-Boolean proof logging revolution! ☺
We're happy to **collaborate**, and **we're hiring**



ACP Summer School '26 128/128

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